(1)

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CSE - 14 (SAP - 1)
25B11CS195
MATHS Assignment

(1) $p\tan x + q\tan y = \tan z$

Sol: A.E: $\frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}$

$$\frac{dx}{\tan x} = \frac{dy}{\tan y}$$

$$\int \frac{dx}{\tan x} = \int \frac{dy}{\tan y}$$

$$\Rightarrow \int \cot x\, dx = \int \cot y\, dy$$

$$\Rightarrow \log(\sin x) = \log(\sin y) + \log c_1$$

$$\Rightarrow \cancel{\log}(\sin x) = \cancel{\log}(c_1 \cdot \sin y)$$

$$\Rightarrow c_1 = \frac{\sin x}{\sin y}$$

$$\frac{dy}{\tan y} = \frac{dz}{\tan z}$$

$$\int \frac{dy}{\tan y} = \int \frac{dz}{\tan z}$$

$$\Rightarrow \int \cot y\, dy = \int \cot z\, dz$$

$$\Rightarrow \log(\sin y) = \log(\sin z) + \log c_2$$

$$\Rightarrow \cancel{\log}(\sin y) = \cancel{\log}(c_2 \cdot \sin z)$$

$$\Rightarrow c_2 = \frac{\sin y}{\sin z}$$

$$\phi(c_1, c_2) = 0$$

$$\Rightarrow \phi\left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z}\right) = 0$$

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② $\frac{y^2 z}{x} p + xzq = y^2$

Sol $\underset{\sim}{A \cdot E}$: $\frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2}$

$\Rightarrow \frac{x\,dx}{y^2 \cancel{z}} = \frac{dy}{x \cancel{z}}$

$\Rightarrow \int x^2 dx = \int y^2 dy$

$\Rightarrow \frac{x^3}{3} = \frac{y^3}{3} + \frac{c_1}{3}$

$\Rightarrow c_1 = x^3 - y^3$

$\frac{x\,dx}{\cancel{y^2} z} = \frac{dz}{\cancel{y^2}}$

$\Rightarrow \int x\,dx = \int z\,dz$

$\Rightarrow \frac{x^2}{2} = \frac{z^2}{2} + \frac{c_2}{2}$

$\Rightarrow c_2 = x^2 - z^2$

$\phi(c_1, c_2) = 0$

$\Rightarrow \phi(x^3 - y^3, x^2 - z^2) = 0$

③ $y^2 p - xyq = x(z - 2y)$

Sol $\underset{\sim}{A \cdot E}$: $\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z-2y)}$

$\Rightarrow \frac{dx}{y\cancel{y}} = \frac{dy}{-x\cancel{y}}$

$\Rightarrow -\int x\,dx = \int y\,dy$

$\Rightarrow -\frac{x^2}{2} = \frac{y^2}{2} - \frac{c_1}{2}$

$\Rightarrow c_1 = x^2 + y^2$

③

$$\frac{dy}{-\cancel{x}y} = \frac{dz}{x(z-2y)}$$

$\Rightarrow (z-2y)\,dy = -y\,dz$

$\Rightarrow z\,dy + y\,dz - 2y\,dy = 0$

$\Rightarrow d(yz) - 2y\,dy = 0$

$\Rightarrow \int d(yz) - 2\int y\,dy = 0$

$\Rightarrow yz - 2\left(\frac{y^2}{2}\right) = c_2$

$\Rightarrow c_2 = yz - y^2$

$\phi(c_1, c_2) = 0$

$\phi\,(x^2+y^2,\ yz - y^2)$

④ $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

Sol: A.E: $\dfrac{dx}{x^2(y-z)} = \dfrac{dy}{y^2(z-x)} = \dfrac{dz}{z^2(x-y)}$

choose $\frac{1}{x^2}, \frac{1}{y^2}, \frac{1}{z^2}$

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x^2}dx + \frac{1}{y^2}dy + \frac{1}{z^2}dz}{\frac{1}{x^2}[x^2(y-z)] + \frac{1}{y^2}[y^2(z-x)] + \frac{1}{z^2}[z^2(x-y)]}$$

$$\Rightarrow \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x^2}dx + \frac{1}{y^2}dy + \frac{1}{z^2}dz}{0}$$

$\Rightarrow \int \frac{1}{x^2}dx + \int \frac{1}{y^2}dy + \int \frac{1}{z^2}dz = 0$

$\Rightarrow -\frac{1}{x} - \frac{1}{y} - \frac{1}{z} = -c_1$

$\Rightarrow c_1 = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$

(4)

choose $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{\frac{1}{x}\left[x^2(y-z)\right] + \frac{1}{y}\left[y^2(z-x)\right] + \frac{1}{z}\left[z^2(x-y)\right]}$$

$$\Rightarrow \frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{xy - xz + yz - yx + zx - zy}$$

$$\Rightarrow \frac{dz}{z^2(x-y)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

$$\Rightarrow \int \frac{1}{x}dx + \int \frac{1}{y}dy + \int \frac{1}{z}dz = 0$$

$$\Rightarrow \log x + \log y + \log z = \log c_2$$

$$\Rightarrow \cancel{\log}(xyz) = \cancel{\log} c_2$$

$$\Rightarrow c_2 = xyz$$

$$\phi(c_1, c_2) = 0$$

$$\phi\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}, xyz\right) = 0$$

(5) $px^2 - qy^2 = z(x-y)$

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A.E : $\frac{dx}{x^2} = \frac{dy}{-y^2} = \frac{dz}{z(x-y)}$

$$\int \frac{dx}{x^2} = \int \frac{dy}{-y^2}$$

$$\Rightarrow \int x^{-2}dx = -\int y^2 dy$$

$$\Rightarrow -\frac{1}{x} = -\left(-\frac{1}{y}\right) - c_1$$

$$\Rightarrow -\frac{1}{x} = \frac{1}{y} - c_1$$

$$\Rightarrow c_1 = \frac{1}{x} + \frac{1}{y}$$

(5)

Choose $\frac{1}{x}, \frac{1}{y}, -\frac{1}{z}$

$$\frac{dx}{x^2} = \frac{dy}{-y^2} = \frac{dz}{z(x-y)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy - \frac{1}{z}dz}{\frac{1}{x}(x^2) - \frac{1}{y}(y^2) - \frac{1}{z}[z(x-y)]}$$

$$\frac{dx}{x^2} = \frac{dy}{-y^2} = \frac{dz}{z(x-y)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy - \frac{1}{z}dz}{x - y - x + y}$$

$$\Rightarrow \frac{dz}{z(x-y)} = \frac{\frac{1}{x}dx + \frac{1}{y}dy - \frac{1}{z}dz}{0}$$

$$\Rightarrow \int \frac{1}{x}dx + \int \frac{1}{y}dy - \int \frac{1}{z}dz = \log c_2$$

$$\Rightarrow \log x + \log y - \log z = \log c_2$$

$$\Rightarrow \log xy - \log z = \log c_2$$

$$\Rightarrow \log\left(\frac{xy}{z}\right) = \log c_2$$

$$\Rightarrow c_2 = \frac{xy}{z}$$

$$\phi(c_1, c_2) = 0$$

$$\phi\left(\frac{1}{x} + \frac{1}{y}, \frac{xy}{z}\right) = 0$$

(6) $(z^2 - 2yz - y^2)p + (xy + zx)q = (xy - zx)$

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A.E : $\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + zx} = \frac{dz}{xy - zx}$

$$\frac{dy}{xy + zx} = \frac{dz}{xy - zx}$$

$$\Rightarrow \frac{dy}{x(y+z)} = \frac{dz}{x(y-z)}$$

$$\Rightarrow (y-z)\,dy = (y+z)\,dz$$

$$\Rightarrow y\,dy - z\,dy = y\,dz + z\,dz$$

$$\Rightarrow \int y\,dy - z\,dz = \int \underline{z\,dy + y\,dz}$$

$$\Rightarrow y\,dy - z\,dz = d(yz)$$

$$\Rightarrow \int y\,dy - \int z\,dz = \int d(yz)$$

$$\Rightarrow \frac{y^2}{2} - \frac{z^2}{2} = yz + c_1$$

$$\Rightarrow c_1 = y^2 - z^2 - 2yz$$

Choose x, y, z

$$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{x(y+z)} = \frac{dz}{x(y-z)} = \frac{x\,dx + y\,dy + z\,dz}{x(z^2 - 2yz - yz) + y[x(y+z)] + z[x(y-z)]}$$

$$\Rightarrow \frac{dz}{x(y-z)} = \frac{x\,dx + y\,dy + z\,dz}{0}$$

$$\Rightarrow x\,dx + y\,dy + z\,dz = 0$$

$$\Rightarrow \int x\,dx + \int y\,dy + \int z\,dz = \frac{c_2}{2}$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{c_2}{2}$$

$$\Rightarrow c_2 = x^2 + y^2 + z^2$$

$$\phi(c_1, c_2) = 0$$

$$\phi(y^2 - z^2 - 2yz,\ x^2 + y^2 + z^2) = 0$$

(7) $(mz - ny)p + (nx - lz)q = ly - mx$

Sol^n^ A.E: $\frac{dx}{mz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx}$

Choose x, y, z

$$\frac{dx}{mz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx} = \frac{x\,dx + y\,dy + z\,dz}{x(mz-ny) + y(nx-lz) + z(ly-mx)}$$

$$\Rightarrow \frac{dz}{ly-mx} = \frac{x\,dx + y\,dy + z\,dz}{0}$$

$$\Rightarrow \int x\,dx + \int y\,dy + \int z\,dz = 0$$

$$\Rightarrow x^2 + y^2 + z^2 = c_1$$

Choose l, m, n

$$\frac{dx}{mz-ny} = \frac{dy}{nx-lz} = \frac{dz}{ly-mx} = \frac{l\,dx + m\,dy + n\,dz}{l(mz-ny) + m(nx-lz) + n(ly-mz)}$$

$$\Rightarrow \frac{dz}{ly-mx} = \frac{l\,dx + m\,dy + n\,dz}{0}$$

$$\Rightarrow \int l\,dx + \int m\,dy + \int n\,dz = 0$$

$$\Rightarrow lx + my + nz = c_2$$

$$\phi(c_1, c_2) = 0$$

$$\phi(x^2 + y^2 + z^2,\ lx + my + nz)$$

(8) $(x^2 - y^2 - z^2)p + 2xyz = 2xz$

Sol^n^ $\frac{dx}{x^2-y^2-z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} \Rightarrow \{A.E\}$

$$\frac{dy}{\cancel{2x}y} = \frac{dz}{\cancel{2x}z}$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{dz}{z}$$

$$\Rightarrow \log y = \log z + \log c_1$$

$$\Rightarrow \log y - \log z = \log c_1$$

$$\Rightarrow \log\left(\frac{y}{z}\right) = \log c_1$$

$$\Rightarrow c_1 = \frac{y}{z}$$

Choose x, y, z

$$\frac{dx}{x^2-y^2-z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} = \frac{x\,dx + y\,dy + z\,dz}{x(x^2-y^2-z^2) + 2xy^2 + 2xz^2}$$

$$\Rightarrow \frac{dz}{2xz} = \frac{x\,dx + y\,dy + z\,dz}{x^3 - xy^2 - xz^2 + 2xy^2 + 2xz^2}$$

$$\Rightarrow \frac{dz}{2xz} = \frac{x\,dx + y\,dy + z\,dz}{x^3 + xy^2 + xz^2}$$

$$\Rightarrow \frac{dz}{2z(\cancel{x})} = \frac{x\,dx + y\,dy + z\,dz}{\cancel{x}(x^2+y^2+z^2)}$$

$$\Rightarrow \int \frac{dz}{z} = \frac{2(x\,dx + y\,dy + z\,dz)}{x^2+y^2+z^2}$$

$$\Rightarrow \int \frac{dz}{z} = \int \frac{d(x^2+y^2+z^2)}{x^2+y^2+z^2}$$

$$\Rightarrow \log z = \log(x^2+y^2+z^2) + c_2$$

$$\Rightarrow \log z - \log(x^2+y^2+z^2) = \log c_2$$

$$\Rightarrow \log\left(\frac{z}{x^2+y^2+z^2}\right) = \log c_2$$

$$\Rightarrow c_2 = \frac{z}{x^2+y^2+z^2}$$

$$\phi(c_1, c_2) = 0$$

$$\phi\left(\frac{y}{z}, \frac{z}{x^2+y^2+z^2}\right) = 0$$

(1)

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Maths Assignment
CSE - 14 (SAP 1)

(9) $4r + 12s + 9t = 0$

$r = \frac{\partial^2 z}{\partial x^2}$, $s = \frac{\partial^2 z}{\partial x \partial y}$, $t = \frac{\partial^2 z}{\partial y^2}$

$\Rightarrow 4 \frac{\partial^2 z}{\partial x^2} + 12 \frac{\partial^2 z}{\partial x \partial y} + 9 \frac{\partial^2 z}{\partial y^2} = 0$

$\because D = \frac{\partial}{\partial x}$ & $D' = \frac{\partial}{\partial y}$ eq^n becomes

$(4D^2 + 12DD' + 9D'^2) z = 0$

$f(D, D') = 4D^2 + 12DD' + 9D'^2$

$Q(x, y) = 0$

A.E $f(M, 1) = 0$

$f(M, 1) \Rightarrow 4M^2 + 12M + 9$

A.E $\Rightarrow 4M^2 + 12M + 9 = 0$

roots : $M = -\frac{3}{2}, -\frac{3}{2}$

C.F $\Rightarrow \phi_1 (y - \frac{3}{2}x) + x\phi_2 (y - \frac{3}{2}x)$

P. I $= 0$

complete sol^n $z = $ C.F

$z = \phi_1 (y - \frac{3}{2}x) + x\phi_2 (y - \frac{3}{2}x)$

(10) $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^3 z}{\partial x^2 \partial y} + 4 \frac{\partial^3 z}{\partial y^3} = e^{x+2y}$

$(D^3 - 3D^2D' + 4D'^3) z = e^{x+2y}$

$f(D, D') = D^3 - 3D^2D' + 4D'^3$

$Q(x, y) = e^{x+2y}$

$f(M, 1) = M^3 - 3M^2 + 4$

A.E $f(M, 1) = 0$

$\Rightarrow M^3 - 3M^2 + 4 = 0$

(2)

roots : $-1, 2$ (repeated 2 times)

C.F $\Rightarrow \phi_1(y-x) + \phi_2(y+2x) + x\phi_3(y+2x)$

$$P.I = \frac{1}{f(D,D')} \cdot Q(x,y)$$

$D = a$
$D = 1$ ✓
$D' = b$
$D' = 2$ ✓

$$P.I = \frac{1}{D^3 - 3D^2D' + 4D'^3} \cdot e^{x+2y}$$

$$P.I = \frac{1}{1-6+32} \cdot e^{x+2y}$$

$$P.I = \frac{1}{27} \cdot e^{x+2y}$$

complete soln $z = C.F + P.I$

$$z = \phi_1(y-x) + \phi_2(y+2x) + x\phi_3(y+2x) + \frac{1}{27} \cdot e^{x+2y}$$

(11) $(D^2 - DD')z = \sin x \cos 2y$

$f(D, D') = D^2 - DD'$

$Q(x,y) = \sin x \cos 2y$

$f(M,1) = M^2 - M$

A.E $f(M,1) = 0$

$M^2 - M = 0$

roots : $1, 0$

C.F $\Rightarrow \phi_1(y+x) + \phi_2(y)$

$$P.I = \frac{1}{f(D,D')} \cdot Q(x,y)$$

$$\cancel{P.I = \frac{1}{D^2 - DD'} \cdot \sin x \cos 2y}$$

$Q(x,y) = \frac{2}{2}\sin x \cos 2y$

$$\cancel{P.I_1 = \frac{1}{D^2 - DD'} \cdot \sin x}$$

$$\underbrace{\frac{\sin(x+2y)}{2}}_{P.I_1} + \underbrace{\frac{\sin(x-2y)}{2}}_{P.I_2}$$

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$$P.I_1 = \frac{1}{f(D,D')} \cdot Q(x,y)$$

$D^2 = -(a)^2$

$D^2 = -1$

$$= \frac{1}{D^2 - DD'} \cdot \frac{\sin(x+2y)}{2}$$

$DD' = -ab$

$DD' = -2$

$$P.I_1 = \frac{1}{-1-(-2)} \cdot \frac{\sin(x+2y)}{2}$$

$D'^2 = -(b)^2$

$D'^2 = -4$

$$P.I_1 = \frac{\sin(x+2y)}{2}$$

$$P.I_2 = \frac{1}{D^2 - DD'} \cdot \frac{\sin(x-2y)}{2}$$

$$P.I_2 = \frac{1}{-1-2} \cdot \frac{\sin(x-2y)}{2}$$

$$P.I_2 = \frac{\sin(x-2y)}{-6}$$

$$P.I = P.I_1 + P.I_2$$

$$P.I = \frac{\sin(x+2y)}{2} - \frac{\sin(x-2y)}{6}$$

complete sol^n $z = C.F + P.I$

$$z = \phi_1(y+x) + \phi_2(y) + \frac{\sin(x+2y)}{2} - \frac{\sin(x-2y)}{6}.$$

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(12) $(D^3 + D^2D' - DD'^2 - D'^3)z = 3\sin(x+y)$

$f(D,D') = D^3 + D^2D' - DD'^2 - D'^3$

$Q(x,y) = 3\sin(x+y)$

$f(M,1) = M^3 + M^2 - M - 1$

A.E : $f(M,1) = 0$

$\Rightarrow M^3 + M^2 - M - 1 = 0$

roots : $1, -1$ (repeated 2 times)

C.F $\Rightarrow \phi_1(y+x) + \phi_2(y-x) + x\phi_3(y-x)$

(4)

$$P.I = \frac{1}{f(D,D')} \cdot Q(x,y)$$

$$P.I = \frac{1}{D^3 + D^2D' - DD'^2 - D'^3} \cdot 3\sin(x+y)$$

$D^2 = -(a)^2$

$D^2 = -1$

$D'^2 = -(b)^2$

$D'^2 = -1$

$DD' = -ab$

$DD' = -1$

$$P.I = \frac{1}{-D - D' + D + D'} \cdot 3\sin(x+y)$$

$$P.I = \infty$$

$$P.I = x \cdot \frac{1}{3D^2 + 2DD' - D'^2} \cdot 3\sin(x+y)$$

$$P.I = x \cdot \frac{1}{-3-2+1} \cdot 3\sin(x+y)$$

$$P.I = -\frac{3}{4} x \sin(x+y)$$

complete sol^n $z = C.F + P.I$

$$z = \phi_1(y+x) + \phi_2(y-x) + x\phi_3(y-x) - \frac{3}{4}\sin(x+y)$$

(13) $$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = x + y$$

$f(D, D') = (D^2 + DD' - 6D'^2)z$ = ~~x+y~~

$Q(x,y) = x + y$

$f(M,1) = M^2 + M - 6$

A.E : $f(M,1) = 0$

~~root~~ $\Rightarrow$ ~~$-3, 2$~~ $\Rightarrow M^2 + M - 6 = 0$

~~C.F~~ $\Rightarrow$ roots : $-3, 2$

C.F $\Rightarrow \phi_1(y - 3x) + \phi_2(y + 2x)$

$$P.I = \frac{1}{f(D,D')} \cdot Q(x,y)$$

$$P.I = \frac{1}{D^2 + DD' - 6D'^2}(x+y)$$

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$$P.I = \frac{1}{D^2\left(\frac{D^2}{D^2} + \frac{D'D}{D^2} - \frac{6D'^2}{D^2}\right)} \cdot (x+y)$$

$$P.I = \frac{1}{D^2\left(1 + \frac{D'}{D} - \frac{6D'^2}{D^2}\right)} \cdot (x+y)$$

$$P.I = \frac{1}{D^2}\left[1 + \underbrace{\left(\frac{D'}{D} - \frac{6D'^2}{D^2}\right)}_{x}\right]^{-1} \cdot (x+y)$$

$$P.I = \frac{1}{D^2}\left[1 - \left(\frac{D'}{D} - \frac{6D'^2}{D^2}\right) + \left(\frac{D'}{D} - \frac{6D'^2}{D^2}\right)_{,} - \dots\right](x+y)$$

$$P.I = \frac{1}{D^2}\left[1 - \frac{D'}{D} + \frac{6D'^2}{D^2}\right](x+y)$$

$$P.I = \frac{1}{D^2}\left[x+y - \frac{D'}{D}(x+y)\right]$$

$$\frac{1}{D} \cdot D'(x+y)$$

$$\Rightarrow \frac{1}{D} \cdot \frac{\partial}{\partial y}(x+y)$$

$$P.I = \frac{1}{D^2}\left[x+y-x\right]$$

$$\Rightarrow \frac{1}{D}(1)$$

$$P.I = \frac{1}{D^2}(y)$$

$$\Rightarrow \int 1\, dx$$

$$\Rightarrow x$$

$$P.I = \int\int y\, dx$$

$$P.I = \int xy\, dx$$

$$P.I = \frac{x^2 y}{2}$$

complete sol^n $z = C.F + P.I$

$$z = \phi_1(y-3x) + \phi_2(y+2x) + \frac{x^2 y}{2}$$

(14) $$\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x} + 3x^2 y$$

$\frac{\partial^3 z}{\partial x^3} = D^3 z$ because ~~differtiating~~ differentiating 3 times w.r.t x.

(6)

$$\frac{\partial^3 z}{\partial x^2 \partial y} = D^2 D' z \quad \left\{ \begin{matrix} 2 \text{ derivatives w.r.t } x \\ 1 \text{ derivative w.r.t } y \end{matrix} \right\}$$

$(D^3 - 2D^2 D')z = 2e^{2x} + 3x^2 y$

$f(D, D') = D^3 - 2D^2 D'$

$Q(x, y) = 2e^{2x} + 3x^2 y$

$f(M, 1) = M^3 - 2M^2$

$\underset{\sim}{A.E}$: $f(M, 1) = 0$

$\Rightarrow M^3 - 2M^2 = 0$

roots: 2, 0 (repeated 2 times)

$\underline{\underline{C.F}} \Rightarrow \phi_1(y + 2x) + \phi_2(y) + x\phi_3(y)$

$$P.I_1 = \frac{1}{f(D, D')} \cdot Q(x, y)$$

$$P.I_1 = \frac{1}{D^3 - 2D^2 D'} \cdot 2e^{2x}$$

$$P.I_1 = \frac{1}{8} \cdot 2e^{2x}$$

$$P.I_1 = \frac{1}{4} \cdot e^{2x}$$

$$P.I_2 = \frac{1}{f(D, D')} \cdot Q(x, y)$$

$$P.I_2 = \frac{1}{D^3 - 2D^2 D'} \cdot 3x^2 y$$

$$P.I_2 = \frac{1}{D^3 \left[\frac{D^3}{D^3} - \frac{2D^2 D'}{D^3} \right]} \cdot 3x^2 y$$

$$P.I_2 = \frac{1}{D^3 \left(1 - \frac{2D'}{D}\right)} \cdot 3x^2 y$$

$$P.I_2 = \frac{1}{D^3} \left[1 - \frac{2D'}{D}\right]^{-1} \cdot 3x^2 y$$

$$P.I_2 = \frac{1}{D^3}\left[1 + \frac{2D'}{D} + \left(\frac{2D'}{D}\right)^2 + \left(\frac{2D'}{D}\right)^3 + \dots\right] 3x^2y$$

$$P.I_2 = \frac{1}{D^3}\left[1 + \frac{2D'}{D} + \frac{4D'^2}{D^2} + \frac{8D'^3}{D^3} + \dots\right] 3x^2y$$

$$P.I_2 = \frac{1}{D^3}\left[3x^2y + 2\frac{D'}{D}(3x^2y)\right]$$

$$P.I_2 = \frac{1}{D^3}\left[3x^2y + 2x^3\right]$$

$$P.I_2 = \int\int\int 3x^2y + 2x^3\, dx\, dx\, dx$$

$$P.I_2 = \int\int 3y\left(\frac{x^3}{3}\right) + 2\left(\frac{x^4}{4}\right) dx\, dx$$

$$P.I_2 = \int y\frac{x^4}{4} + \frac{1}{2}\cdot\frac{x^5}{5}\, dx$$

$$P.I_2 = \frac{y}{4}\left(\frac{x^5}{5}\right) + \frac{1}{10}\left(\frac{x^6}{6}\right)$$

$$P.I_2 = \frac{x^5y}{20} + \frac{x^6}{60}$$

$$2\frac{D'}{D}(3x^2y)$$

$$\Rightarrow 2\left(\frac{1}{D}\right)\frac{\partial}{\partial y}(3x^2y)$$

$$\Rightarrow 2\left(\frac{1}{D}\right)(3x^2)$$

$$\Rightarrow \frac{1}{D}(6x^2)$$

$$\Rightarrow 6\int x^2\, dx$$

$$\Rightarrow 2x$$

$$P.I = P.I_1 + P.I_2$$

$$P.I = \frac{1}{4}\cdot e^{2x} + \frac{x^5y}{20} + \frac{x^6}{60}$$

complete sol^n $z = C.F + P.I$

$$z = \phi_1(y) + \phi_2(y+2x) + x\phi_3(y+2x) + \frac{1}{4}\cdot e^{2x} + \frac{x^5y}{20} + \frac{x^6}{60}$$

(8)

(15) $(D^2 - DD' - 2D'^2)z = (y-1)e^x$

$f(D, D') = D^2 - DD' - 2D'^2$

$Q(x, y) = (y-1)e^x$ product function

A.E $f(M, 1) = 0$

$\Rightarrow M^2 - M - 2 = 0$

roots : $-1, 2$ $\boxed{\int fg\,dx - \int \frac{d}{dx}(f)\{\int g\,dx\}\,dx}$

$C.F \Rightarrow \phi_1(y-x) + \phi_2(y+2x)$

$P.I = \frac{1}{f(D,D')} \cdot Q(x,y)$

$= \frac{1}{D^2 - DD' - 2D'^2} \cdot (y-1)e^x$

$= \frac{1}{[D-(2D')][D-(-1D')]} \cdot (y-1)e^x$

$= \frac{1}{D+D'} \cdot \frac{1}{D-2D'} \cdot (y-1)e^x$ $\quad y = c + mx$, $y = c - 2x$, $c = y + 2x$

$= \frac{1}{D+D'} \int (c - 2x - 1)e^x\,dx$

$= \frac{1}{D+D'} [(c-2x-1)\int e^x\,dx - \int \frac{d}{dx}(c-2x-1)(\int e^x\,dx)\,dx$

$= \frac{1}{D+D'} [(c-2x-1)e^x - \int (-2)e^x\,dx$

$= \frac{1}{D+D'} [(c-2x-1)e^x + 2e^x$

$= \frac{1}{D+D'} [(y+2x-2x-1)e^x + 2e^x]$

$= \frac{1}{D+D'} \cdot (y-1)e^x + 2e^x$

$= \frac{1}{D+D'} \cdot (y-1+2)e^x$ $\quad y = c + mx$, $y = c + x$, $c = y - x$

$= \frac{1}{D+D'} \cdot (y+1)e^x$

$= \int (c + x + 1)\, e^x \, dx$

$\;\;\;\;\;\downarrow\;\;\;\;\;\;\;\downarrow$

$\;\;\;\;\;\text{fun}\;\;\;\;\text{fun}$

$= (c+x+1)\int e^x dx - \int \frac{d}{dx}(c+x+1)\left\{\int e^x dx\right\} dx$

$= (c+x+1)e^x - \int 1\, e^x dx$

$= (c+x+1)\, e^x - e^x$

$= (y - x + x + 1)\, e^x - e^x$

$= (y+1)\, e^x - e^x$

$= (y+1-1)\, e^x$

$= y e^x$

$P.I = y e^x$

complete sol^n $z = C.F + P.I$

$z = \phi_1(y+2x) + \phi_2(y-x) + y e^x$

⑥ $(D^2 + 2DD' - 8D'^2)\, z = \sqrt{2x+3y}$

$f(D, D') = D^2 + 2DD' - 8D'^2$

$Q(x, y) = \sqrt{2x+3y}$

A.E $f(M, 1) = 0$

$\Rightarrow M^2 + 2M - 8 = 0$

roots : $2, -4$

$C.F \Rightarrow \phi_1(y+2x) + \phi_2(y-4x)$

$P.I = \frac{1}{f(D,D')} \cdot Q(x,y)$

$P.I = \frac{1}{D^2 + 2DD' - 8D'^2} \cdot \sqrt{2x+3y}$

$= \frac{1}{[D-(-2D')][D-(-4D')]} \cdot \sqrt{2x+3y}$

(10)

$$= \frac{1}{(D-2D')(D+4D')} \cdot \sqrt{2x+3y}$$

$$= \frac{1}{D-2D'} \cdot \frac{1}{D+4D'} \cdot \sqrt{2x+3y}$$

$y = c + mx$

$y = c + 4x$

$c = y - 4x$

$$= \frac{1}{D-2D'} \int \sqrt{2x + 3(c+4x)}\, dx$$

$$= \frac{1}{D-2D'} \int \sqrt{2x + 3c + 12x}\, dx$$

$$= \frac{1}{D-2D'} \int \sqrt{3c + 14x}\, dx$$

$3\overset{\text{const}}{\uparrow}c + 14x = t$

$14\,dx = dt$

$$= \frac{1}{D-2D'} \int \frac{\sqrt{t}}{14}\, dt$$

$$= \frac{1}{D-2D'} \cdot \frac{1}{14} \cdot \frac{t^{1/2+1}}{1/2+1}$$

$$= \frac{1}{D-2D'} \cdot \frac{1}{14} \cdot \frac{2}{3} t^{3/2}$$

$$= \frac{1}{21} \cdot \frac{1}{D-2D'} \cdot (3c+14x)^{3/2}$$

$$= \frac{1}{21} \cdot \frac{1}{D-2D'} [3y - 12x + 14x]^{3/2}$$

$$= \frac{1}{21} \cdot \frac{1}{D-2D'} [3y + 2x]^{3/2}$$

$y = c - 2x$

$c = y + 2x$

$$= \frac{1}{21} \int (3y + 2x)^{3/2} dx$$

$$= \frac{1}{21} \int [3(c-2x) + 2x]^{3/2} dx$$

$3c - 4x = t$

$-4\,dx = dt$

$dx = \frac{dt}{-4}$

$$= \frac{1}{21} \int (3c - 6x + 2x)^{3/2} dx$$

$$= \frac{1}{21} \int (3c - 4x)^{3/2} dx$$

$$= \frac{1}{21} \left(-\frac{1}{4}\right) \int t^{3/2} dt$$

$$= \frac{1}{21(-4)} \cdot \frac{t^{5/2}}{5/2}$$

$$= \frac{1}{21(-10)} \cdot t^{5/2}$$

$$= -\frac{1}{210} (3c - 4x)^{5/2}$$

$$= -\frac{1}{210} (3(y+2x) - 4x))^{5/2}$$

$$= -\frac{1}{210} [3y + 6x - 4x]^{5/2}$$

$$= -\frac{1}{210} [3y + 2x]^{5/2}$$

Complete sol^n $z = C.F + P.I$

$$z = \phi_1 (y + 2x) + \phi_2 (y - 4x) - \frac{1}{210} [3y + 2x]^{5/2}$$

①

Assignment-4

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AIML-10

1) Find the angle b/w the surfaces $x^2+y^2+z^2=9$ and $z=x^2+y^2-3$ at $(2,-1,2)$.

A) given $f = x^2+y^2+z^2-9$

$g = x^2+y^2-z-3$

Now

$\frac{\partial f}{\partial x} = 2x \quad \frac{\partial f}{\partial y} = 2y \quad \frac{\partial f}{\partial z} = -9$

$\nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$

$\nabla f = \bar{i}(2x) + \bar{j}(2y) + \bar{k}(2z)$

Now

$\frac{\partial g}{\partial x} = 2x, \quad \frac{\partial g}{\partial y} = 2y; \quad \frac{\partial g}{\partial z} = -1$

$\nabla g = \bar{i}\frac{\partial g}{\partial x} + \bar{j}\frac{\partial g}{\partial y} + \bar{k}\frac{\partial g}{\partial z}$

$\nabla g = \bar{i}(2x) + \bar{j}(2y) + \bar{k}(-1)$

$\nabla g = 2x\bar{i} + 2y\bar{j} - \bar{k}$

Let $\overline{n_1}$ & $\overline{n_2}$ be the normals to the surfaces f & g at $(2,-1,2)$ respectively

$(\nabla f)_{(2,-1,2)} = 2(2)\bar{i} + 2(-1)\bar{j} + \bar{k}(2(2))$

$= 4\bar{i} - 2\bar{j} + 4\bar{k} \Rightarrow \overline{n_1}$

$(\nabla g)_{(2,-1,2)} = 2(2)\bar{i} + 2(-1)\bar{j} - \bar{k}$

$= 4\bar{i} - 2\bar{j} - \bar{k} \Rightarrow \overline{n_2}$

Let θ be the angle b/w f & g

$$\cos\theta = \frac{\overline{n_1}\cdot\overline{n_2}}{|\overline{n_1}|\cdot|\overline{n_2}|}$$

$$= \frac{(4\bar{i}-2\bar{j}+4\bar{k})\cdot(4\bar{i}-2\bar{j}-\bar{k})}{\sqrt{16+4+16}\ \sqrt{16+4+1}}$$

$$= \frac{16+4-4}{\sqrt{36}\ \sqrt{21}}$$

$$= \frac{\cancel{16}\,8}{\cancel{6}_{3}\sqrt{9\times3}}$$

$$\cos\theta = \frac{8}{3\sqrt{21}}$$

$$\boxed{\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)}$$

2) Find the angle b/w the surfaces $xy = z^2$ at $(4,1,2)$ & $(3,3,-3)$.

A) Let $f = xy - z^2$

Now

$$\frac{\partial f}{\partial x} = y,\ \frac{\partial f}{\partial y} = x,\ \frac{\partial f}{\partial z} = -2z$$

$$\nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$$

$$\nabla f = \bar{i}(y) + \bar{j}(x) + \bar{k}(-2z)$$

$$\nabla f = y\bar{i} + x\bar{j} - 2z\bar{k}$$

Let $\overline{n_1}$ & $\overline{n_2}$ be the normal to the surface f at $(4,1,2)$ & $(3,3,-3)$

$\bar{n}_1 = (\nabla f)_{(4,1,2)} = (1)\bar{i} + (4)\bar{j} - 2(2)\bar{k}$

$\bar{n}_1 = \bar{i} + 4\bar{j} - 4\bar{k}$

$\bar{n}_2 = (\nabla f)_{(3,3,-3)} = (3)\bar{i} + (3)\bar{j} - 2(-3)\bar{k}$

$\bar{n}_2 = 3\bar{i} + 3\bar{j} + 6\bar{k}$

Let θ be the angle b/w $\bar{n}_1$ & $\bar{n}_2$

$$\cos\theta = \frac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1| \cdot |\bar{n}_1|}$$

$$= \frac{(\bar{i}+4\bar{j}-4\bar{k})(3\bar{i}+3\bar{j}+6\bar{k})}{\sqrt{1+16+16}\sqrt{9+9+36}}$$

$$= \frac{3+12-24}{\sqrt{33}\sqrt{54}}$$

$$= \frac{-9}{\sqrt{33}\sqrt{9\times 6}}$$

$$= \frac{-\cancel{9}\,3}{\cancel{3}\sqrt{6}\sqrt{33}}$$

$$\cos\theta = \frac{-3}{\sqrt{6}\sqrt{33}}$$

$$\boxed{\theta = \cos^{-1}\left(\frac{-3}{\sqrt{6}\sqrt{33}}\right)}$$

3) Find the values of a & b such that the surfaces $ax^2 - byz = (a+2)x$ & $4x^2y + z^3 = 4$ cut orthogonally at $(1,-1,2)$

A) given surfaces are

$$ax^2 - byz = (a+2)x \rightarrow ①$$
$$4x^2y + z^3 = 4 \rightarrow ②$$

given that ① & ② intersects at point $(1,-1,2)$
hence the point $(1,-1,2)$ satisfies both the equations

From ① we have

$$a(1)^2 - b(-1)(2) = (a+2)(1)$$
$$\cancel{a + bz = a + 2}$$
$$\cancel{a} + 2b = \cancel{a} + 2$$
$$\boxed{b = 1}$$

Let $f = ax^2 - byz - (a+2)x$

$$f = ax^2 - yz - (a+2)x \qquad (\because b = 1)$$

& $g = 4x^2y + z^3 - 4$

$$\nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$$

$$\cancel{\nabla f = \bar{i}(2ax) + \bar{j}(-z) + \bar{k}(}$$

$$\frac{\partial f}{\partial x} = 2ax - (a+2) \quad \frac{\partial f}{\partial y} = -z \ , \ \frac{\partial f}{\partial z} = -y$$

$$\nabla f = \bar{i}(2ax - (a+2)) + \bar{j}(-z) + \bar{k}(-y)$$

$$\nabla g = \bar{i}\frac{\partial g}{\partial x} + \bar{j}\frac{\partial g}{\partial y} + \bar{k}\frac{\partial g}{\partial z}$$

$$\frac{\partial g}{\partial x} = 8xy \ , \ \frac{\partial g}{\partial y} = 4x^2 \ , \ \frac{\partial g}{\partial z} = 3z^2$$

$\nabla g = \bar{i}(8xy) + \bar{j}(4x^2) + \bar{k}(3z^2)$

Let $\bar{n_1}$ & $\bar{n_2}$ be the normal to the surface of f & g at $(1,-1,2)$ respectively.

$\therefore \bar{n_1} = (\nabla f)_{(1,-1,2)} = (a-2)\bar{i} - 2\bar{j} + \bar{k}$

$\bar{n_2} = (\nabla g)_{(1,-1,2)} = -8\bar{i} + 4\bar{j} + 12\bar{k}$

Since (1) & (2) intersects orthogonally.

$\therefore \bar{n_1} \cdot \bar{n_2} = 0$

$((a-2)\bar{i} - 2\bar{j} + \bar{k}) \cdot (-8\bar{i} + 4\bar{j} + 12\bar{k}) = 0$

$(a-2)(-8) - 2(4) + 12 = 0$

$-8a + 16 - 8 + 12 = 0$

$8a = 28 - 8$

$8a = 20$

$a = \cancel{20}^{5} / \cancel{8}_{2}$

$\boxed{a = 5/2}$

4) find the directional derivative of $f(x,y,z) = xy^3 + yz^3$ at $(2,-1,1)$ in the direction of the vector $\bar{i} + 2\bar{j} + 2\bar{k}$.

A) given $f(x,y,z) = xy^3 + yz^3$

Now

$\frac{\partial f}{\partial x} = y^3$, $\frac{\partial f}{\partial y} = 3xy^2 + z^3$, $\frac{\partial f}{\partial z} = 3yz^2$

$$\nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$$

$$\nabla f = \bar{i}(y^3) + \bar{j}(3yx^2 + z^3) + \bar{k}(3yz^2)$$

$$\therefore (\nabla f)_{(2,-1,1)} = \bar{i}(-1)^3 + \bar{j}(3(-1)(2)^2 + (1)^3) + \bar{k}(3(-1)(1))$$

$$= -\bar{i} + \bar{j}(-12+1) + \bar{k}(-3)$$

$$\therefore (\nabla f)_{(2,-1,1)} = -\bar{i} - 11\bar{j} - 3\bar{k}$$

Let $\bar{e}$ be the unit vector in the directoon of the vector

$$\bar{a} = \bar{i} + 2\bar{j} + 2\bar{k}$$

$$\therefore \bar{e} = \frac{\bar{a}}{|\bar{a}|} = \frac{\bar{i}+2\bar{j}+2\bar{k}}{\sqrt{1+4+4}} = \frac{1}{3}(\bar{i}+2\bar{j}+2\bar{k})$$

$$\therefore D\cdot D = \nabla f \cdot \bar{e}$$

$$= (-\bar{i} - 11\bar{j} - 3\bar{k}) \cdot \frac{1}{3}(\bar{i}+2\bar{j}+2\bar{k})$$

$$= \frac{1}{3}[-1-22-6]$$

$$D\cdot D = -27/3$$

5) find the directional derivative of $\phi(x,y,z) = xy^2 + yz^3$ ~~at (2,-1,1) in the direction of~~ normal to the surface

(4) $x\log z - y^2 = -4$ at $(-1,2,1)$

A) given that $\phi(x,y,z) = xy^2 + yz^3$

Now

$$\frac{\partial \phi}{\partial x} = y^2, \quad \frac{\partial \phi}{\partial y} = 2xy + z^3, \quad \frac{\partial \phi}{\partial z} = 3yz^2$$

$$\text{grad}\,\phi = \bar{i}\frac{\partial \phi}{\partial x} + \bar{j}\frac{\partial \phi}{\partial y} + \bar{k}\frac{\partial \phi}{\partial z}$$

$$\nabla\phi = \bar{i}(y^2) + \bar{j}(2yx + z^3) + \bar{k}(3z^2 y)$$

$$(\nabla\phi)_{(2,-1,1)} = \bar{i} - 3\bar{j} - 3\bar{k}$$

Let $f = x\log z - y^2 + 4$

<u>Now</u>

$$\frac{\partial f}{\partial x} = \log z,\ \frac{\partial f}{\partial y} = -2y,\ \frac{\partial f}{\partial z} = \frac{x}{z}$$

$$\therefore \nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$$

$$= \bar{i}(\log z) + \bar{j}(-2y) + \bar{k}\left(\frac{x}{z}\right)$$

$$= \bar{i}(\log z) + \bar{j}(-2y) + \left(\frac{x}{z}\right)\bar{k}$$

Let $\bar{n}$ be normal to the surface f at $(-1,2,1)$

$$\bar{n} = (\nabla f)_{(-1,2,1)}$$

$$\bar{n} = -4\bar{j} - \bar{k}$$

$\therefore$ unit normal vector

$$\bar{e} = \frac{\bar{n}}{|\bar{n}|} = \frac{1}{\sqrt{17}}(-4\bar{j} - \bar{k})$$

$$\therefore D \cdot D = \nabla\phi \cdot \bar{e} = (\bar{i} - 3\bar{j} - 3\bar{k})\frac{1}{\sqrt{17}}(-4\bar{j} - \bar{k})$$

$$= \frac{1}{\sqrt{17}}(12+3)$$

$$= \frac{15}{\sqrt{17}}$$

6) Find the directional derivative of $x^2 - y^2 + 2z^2$ at the point $P(1,2,3)$ in the direction of the line PQ where Q is the point $(5,0,4)$. Also find the magnitude of the maximum directional derivative.

A) given that $f = x^2 - y^2 + 2z^2$

Now

$\frac{\partial f}{\partial x} = 2x$, $\frac{\partial f}{\partial y} = -2y$, $\frac{\partial f}{\partial z} = 4z$

$\nabla f = \bar{i}\frac{\partial f}{\partial x} + \bar{j}\frac{\partial f}{\partial y} + \bar{k}\frac{\partial f}{\partial z}$

$\nabla f = \bar{i}(2x) + \bar{j}(-2y) + \bar{k}(4z)$

$\nabla f = 2x\bar{i} - 2y\bar{j} + 4z\bar{k}$

$\therefore (\nabla f)_{(1,2,3)} = 2\bar{i} - 4\bar{j} + 12\bar{k}$

Let $\overline{OP}$ & $\overline{OQ}$ be the position vectors of the points $P(1,2,3)$ & $Q(5,0,4)$ w.r.t origin

$\overline{OP} = \bar{i} + 2\bar{j} + 3\bar{k}$

$\overline{OQ} = 5\bar{i} + 4\bar{k}$

$\overline{PQ} = \overline{OP} - \overline{OQ}$

$= \bar{i} + 2\bar{j} + 3\bar{k} - 5\bar{i} - 4\bar{k}$

$\overline{PQ} = -4\bar{i} + 2\bar{j} - \bar{k}$

Let $\bar{e}$ be the unit vector in the direction of the line $\overline{PQ}$

$\therefore \bar{e} = \frac{\overline{PQ}}{|\overline{PQ}|} = \frac{1}{\sqrt{21}}(4\bar{i} - 2\bar{j} - \bar{k})$

$\therefore D \cdot D = \nabla f \cdot \bar{e} = (2\bar{i} - 4\bar{j} + 12\bar{k})\frac{1}{\sqrt{21}}(4\bar{i} - 2\bar{j} + \bar{k})$

$= \frac{1}{\sqrt{21}}(8 + 8 + 12)$

$D \cdot D = 28/\sqrt{21}$

$\therefore$ maximum value $= |\nabla f| = \sqrt{2^2 + (-4)^2 + 6^2} = \sqrt{56} = 2\sqrt{14}$

7) Find the directional derivative of $xy^2+yz^2+zx^2$ along the tangent to the curve $x=t, y=t^2, z=t^3$ at the point $(1,1,1)$

A) given $\phi = xy^2+yz^2+zx^2$

Now

$\frac{\partial\phi}{\partial x} = y^2+2zx$, $\frac{\partial\phi}{\partial y} = 2xy+z^2$, $\frac{\partial\phi}{\partial z} = 2yz+x^2$

$\text{grad}\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$

$\nabla\phi = \bar{i}(y^2+2zx)+\bar{j}(2xy+z^2)+\bar{k}(2yz+x^2)$

$(\nabla\phi)_{(1,1,1)} = 3\bar{i}+3\bar{j}+3\bar{k}$

Let $\bar{r} = x\bar{i}+y\bar{j}+z\bar{k}$ be the position vector of any point $P(x,y,z)$ on the curve

$x=t, y=t^2, z=t^3$

$\therefore \bar{r} = t\bar{i}+t^2\bar{j}+t^3\bar{k}$

$\frac{\partial\bar{r}}{\partial t} = \bar{i}+2t\bar{j}+3t^2\bar{k}$

at $x=1, y=1, z=1$ We have $t=1$

$\therefore$ Tangent vector at $(1,1,1)$ is given by

$\left(\frac{\partial\bar{r}}{\partial t}\right)_{t=1} = \bar{i}+2\bar{j}+3\bar{k}$

$\therefore$ unit tangent vector

$\bar{e} = \frac{\bar{i}+2\bar{j}+3\bar{k}}{\sqrt{1+4+9}} = \frac{1}{\sqrt{14}}(\bar{i}+2\bar{j}+3\bar{k})$

$\therefore D\cdot D = \nabla\phi\cdot\bar{e}$

$= (3\bar{i}+3\bar{j}+3\bar{k})\cdot\frac{1}{\sqrt{14}}(\bar{i}+2\bar{j}+3\bar{k})$

$= \frac{1}{\sqrt{14}}(3+6+9)$

$D\cdot D = \frac{18}{\sqrt{14}}$

8) The temperature of points in a space is given by $T(x,y,z) = x^2+y^2-z$, a mosquito is located at $(1,1,2)$ desires to fly in such a direction that it will get warm as soon as possible. In what direction should it move. Also find what is maximum rate of change in the temperature.

A) given $T(x,y,z) = x^2+y^2-z$

Now

$\frac{\partial T}{\partial x} = 2x, \ \frac{\partial T}{\partial y} = 2y, \ \frac{\partial T}{\partial z} = -1$

$\nabla T = \bar{i}\frac{\partial T}{\partial x} + \bar{j}\frac{\partial T}{\partial y} + \bar{k}\frac{\partial T}{\partial z}$

$\nabla T = \bar{i}(2x) + \bar{j}(2y) + \bar{k}(-1)$

$(\nabla T)_{(1,1,2)} = \bar{i}(2(1)) + \bar{j}(2(1)) + \bar{k}(-1)$

$(\nabla T)_{(1,1,2)} = 2\bar{i}+2\bar{j}-\bar{k}$

Let $\bar{n}$ be the normal vector to the surface T at $(1,1,2)$

$\bar{n} = \nabla\phi(x,y,z)$ at given poiven

$\bar{n} = 2(1)\bar{i} + 2(1)\bar{j} - \bar{k}$

$\bar{n} = 2\bar{i} + 2\bar{j} - \bar{k}$

$\therefore$ The mosquito should move in the direction of the vector $(2,2,-1)$

$\therefore$ The maximum rate of change in temperature

$$|\nabla T| = \sqrt{2^2+2^2+(-1)^2} = \sqrt{4+4+1}$$
$$= \sqrt{9}$$
$$= 3 \text{ units}$$

$\therefore$ Direction $= 2\bar{i} + 2\bar{j} - \bar{k}$

9) Find $\text{div}\,\bar{F}$ & $\text{curl}\,\bar{F}$, given that $\bar{F} = \text{grad}(x^3+y^3+z^3-3xyz)$

10)

A) given that

$$\bar{F} = \text{grad}(x^3+y^3+z^3-3xyz)$$

Let $\phi = x^3+y^3+z^3-3xyz$

$$\therefore \nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$F = \bar{i}(3x^2-3yz) + \bar{j}(3y^2-3xz) + \bar{k}(3z^2-3xy)$$

$$= F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$$

$$\therefore \text{div}\bar{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$\text{div}\bar{F} = 6x + 6y + 6z$$

$$\therefore \text{curl}\,\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

$$= \begin{vmatrix} i & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2-3yz & 3y^2-3xz & 3z^2-3xy \end{vmatrix}$$

$$= \bar{i}\left[\frac{\partial}{\partial y}(3z^2-3xy) - \frac{\partial}{\partial z}(3y^2-3xz)\right] - \bar{j}\left[\frac{\partial}{\partial x}(3z^2-3xy) - \frac{\partial}{\partial z}(3x^2-3yz)\right] + \bar{k}\left[\frac{\partial}{\partial x}(3y^2-3xz) - \frac{\partial}{\partial y}(3x^2-3yz)\right]$$

$$= \bar{i}[-3x+3x] - \bar{j}[-3y+3y] + \bar{k}[-3z+3z]$$

$$= \bar{0}$$

$$\therefore \text{curl}(\text{grad}\,\phi) = \bar{0}$$

10) find $\operatorname{div}\bar{F}$ & $\operatorname{curl}\bar{F}$ at $(1,-1,1)$, given that
$\bar{F} = xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}$

A) given $\bar{F} = xy^2\bar{i} + 2x^2yz\bar{j} - 3yz^2\bar{k}$

$$= f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$$

$$\therefore \operatorname{div}\bar{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$\operatorname{div}\bar{F} = y^2 + 2x^2z - 6yz$$

$\operatorname{div}\bar{F}$ at $(1,-1,1)$

$$\operatorname{div}\bar{F} = (-1)^2 + 2(1)^2(1) - 6(-1)(1)$$

$$= 1 + 2 + 6$$

$$\operatorname{div}\bar{F} = 9$$

$$\operatorname{curl}\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & 2x^2yz & -3yz^2 \end{vmatrix}$$

$$= \bar{i}\left[\frac{\partial}{\partial y}(-3yz^2) - \frac{\partial}{\partial z}(2x^2yz)\right] - \bar{j}\left[\frac{\partial}{\partial x}(-3yz^2) - \frac{\partial}{\partial z}(xy^2)\right]$$
$$+ \bar{k}\left[\frac{\partial}{\partial x}(2x^2yz) - \frac{\partial}{\partial y}(xy^2)\right]$$

$$\operatorname{curl}\bar{F} = \bar{i}\left[-3z^2 - 2x^2y\right] - \bar{j}\left[0\right] + \bar{k}\left[4xyz - 2xy\right]$$

$\operatorname{curl}\bar{F}$ at $(1,-1,1)$

$\cancel{\text{Curl}\,\bar{F} = \bar{i}\,[4(1)(-1)(1) - 2}$

$$\text{Curl}\,\bar{F} = \bar{i}\,[-3(1)^2 - 2(1)^2(-1)] + \bar{k}\,[4(1)(-1)(1) - 2(1)(-1)]$$

$$= \bar{i}\,[-3+2] + \bar{k}\,[-4+2]$$

$$\text{Curl}\,\bar{F} = -\bar{i} - 2\bar{k}$$

---

11) P.T $\text{div}(r^n \bar{r}) = (n+3)r^n$, hence show that $\frac{\bar{r}}{r^3}$ is solenoidal

A) Let $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$$r = |\bar{r}| = \sqrt{x^2+y^2+z^2}$$

$$r^2 = x^2+y^2+z^2 \rightarrow ①$$

diff. ① w.r.t 'x' partially

$$\cancel{2}\, r \frac{dr}{dx} = \cancel{2}\, x$$

$$\frac{dr}{dx} = \frac{x}{r}$$

$|||^{ly}$

$$\frac{dr}{dy} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

Let $\bar{F} = r^n \bar{r}$

$$= r^n(x\bar{i} + y\bar{j} + z\bar{k})$$

$$= x r^n \bar{i} + y r^n \bar{j} + z r^n \bar{k}$$

$$= f_1 \bar{i} + f_2 \bar{j} + f_3 \bar{k} \quad (\because \text{say})$$

$$\therefore \text{div}\bar{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$\text{div}(r^n \bar{r}) = \frac{\partial}{\partial x}(x r^n) + \frac{\partial}{\partial y}(y r^n) + \frac{\partial}{\partial z}(z r^n)$$

b) $\operatorname{div} r^n \bar{r} = x n r^{n-1}\frac{\partial r}{\partial x} + r^n \cdot 1 + y n r^{n-1}\frac{\partial r}{\partial y} + r^n \cdot 1$
$+ z r^{n-1}\frac{\partial r}{\partial z} + r^n \cdot 1$

$= 3r^n + n r^{n-1}\left[x\frac{\partial r}{\partial x} + y\frac{\partial r}{\partial y} + z\frac{\partial r}{\partial z}\right]$

$= 3r^n + n r^{n-1}\left[\frac{x^2}{r} + \frac{y^2}{r} + \frac{z^2}{r}\right]$

$= 3r^n + n r^{n-1}\left(\frac{x^2+y^2+z^2}{r}\right)$

$= 3r^n + n r^{n-1}\left(\frac{r^2}{r}\right)$

$= 3r^n + n r^n$

$\therefore \operatorname{div} r^n \bar{r} = (3+n) r^n$

Now Sub $n = -3$

$\operatorname{div}(r^{-3}\bar{r}) = (3-3) r^{-3}$

$\operatorname{div}\left(\frac{\bar{r}}{r^3}\right) = (3-3) r^{-3}$

$\operatorname{div}\left(\frac{\bar{r}}{r^3}\right) = 0$

So, $\frac{\bar{r}}{r^3}$ is solenoidal.

12) show that the vector field $\bar{F} = 2xyz^2\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$ is irrotational & find the potential function.

A) given $\bar{F} = 2xyz^2\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$

$= f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$

$$\text{Curl } \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz^2 & x^2z^2 + z\cos yz & 2x^2yz + y\cos yz \end{vmatrix}$$

$$= \bar{i}\left[\frac{\partial}{\partial y}(2x^2yz + y\cos yz) - \frac{\partial}{\partial z}(x^2z^2 + z\cos yz)\right]$$
$$-\bar{j}\left[\frac{\partial}{\partial x}(2x^2yz + y\cos yz) - \frac{\partial}{\partial z}(2xyz^2)\right]$$
$$+\bar{k}\left[\frac{\partial}{\partial x}(x^2z^2 + z\cos yz) - \frac{\partial}{\partial y}(2xyz^2)\right]$$

$$= \bar{i}\left[2x^2z + y(-z\sin yz) + \cos yz\right] - \frac{\partial}{\partial z}\left(2x^2z + z(-\sin yz) + \cos yz\right)\Big]$$
$$-\bar{j}\left[4xyz - 4xyz\right] + \bar{k}\left[2xz^2 - 2xz^2\right]$$

$$= \bar{0}$$

$\text{Curl } \bar{F} = 0$

hence $\bar{F}$ is irrotational

Since F irrotational hence their exists of scalar potential function ϕ such that $\bar{F} = \nabla\phi$

$$\bar{F} = \nabla\phi$$

$$(2xyz^2)\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$$
$$= \frac{\partial\phi}{\partial x}\bar{i} + \frac{\partial\phi}{\partial y}\bar{j} + \frac{\partial\phi}{\partial z}\bar{k}$$

$\therefore \frac{\partial \phi}{\partial x} = 2xyz^2 \xrightarrow[(y,z \text{ as cont.})]{\text{I.O.B w.r.t } x} 2\frac{x^2}{2} yz^2 \Rightarrow x^2yz^2 + g_1(y,z)$

$\therefore \frac{\partial \phi}{\partial y} = x^2z^2 + z\cos yz \xrightarrow[(x,z \text{ as cont.})]{\text{I.O.B w.r.t } y} z\left(\frac{\sin yz}{z}\right) + x^2z^2y$

$\Rightarrow x^2z^2y + \sin yz + g_2(x,z)$

$\therefore \frac{\partial \phi}{\partial z} = 2x^2yz + y\cos yz \xrightarrow[(y,x \text{ as cont.})]{\text{I.O.B w.r.t } z} 2x^2y\frac{z^2}{2} + y\left(\frac{\sin yz}{y}\right)$

$\Rightarrow x^2yz^2 + \sin yz + g_3(x,y)$

$\therefore \phi = x^2yz^2 + \sin yz$

13) Show that the vector field $\bar{F} = (x^2-yz)\bar{i} + (y^2-zx)\bar{j} + (z^2-xy)\bar{k}$ is irrotational & hence find the scalar potential function corresponding to it

A) given $\bar{F} = (x^2-yz)\bar{i} + (y^2-zx)\bar{j} + (z^2-xy).\bar{k}$

$= f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$

$$\text{Curl}\,\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2-yz & y^2-zx & z^2-xy \end{vmatrix}$$

$$= \bar{i}\left[\frac{\partial}{\partial y}(z^2-xy) - \frac{\partial}{\partial z}(y^2-zx)\right] - \bar{j}\left[\frac{\partial}{\partial x}(z^2-xy) - \frac{\partial}{\partial z}(x^2-yz)\right] + \bar{k}\left[\frac{\partial}{\partial x}(y^2-zx) - \frac{\partial}{\partial y}(x^2-yz)\right]$$

$$= \bar{i}[-x+x] - \bar{j}[-y+y] + \bar{k}[-z+z]$$

$\text{Curl } \bar{F} = \bar{0}$

hence $\bar{F}$ is irrotational

Since $\bar{F}$ is irrotational hence their exist a scalar potential function ϕ such that $\bar{F} = \nabla\phi$

$$\bar{F} = \nabla\phi$$

$$(x^2 - yz)\bar{i} + (y^2 - zx)\bar{j} + (z^2 - xy)\bar{k} = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$\therefore \frac{\partial\phi}{\partial x} = x^2 - yz \xrightarrow[(y,z \text{ as const.})]{\text{I.O.B w.r.t }(x)} \frac{x^3}{3} - xyz + g_1(y,z)$$

$$\therefore \frac{\partial\phi}{\partial y} = y^2 - zx \xrightarrow[(x,z \text{ as const})]{\text{I.O.B w.r.t }(y)} \frac{y^3}{3} - xyz + g_2(x,z)$$

$$\therefore \frac{\partial\phi}{\partial z} = z^2 - xy \xrightarrow[(y,\cancel{z} \text{ as const.})]{\text{I.O.B w.r.t }(z)} \frac{z^3}{3} - xyz + g_3(x,y)$$

$$\therefore \phi = \frac{x^3}{3} - xyz + \frac{y^3}{3} + \frac{z^3}{3}$$

$$\therefore \phi = \frac{1}{3}(x^3 + y^3 + z^3) - xyz$$

14) Find the constants a, b, c so that the following vector $\bar{A}$ is irrotational. Also find its scalar potential function. given that $\bar{A} = (x + 2y + az)\bar{i} + (bx - 3y - z)\bar{j} + \cancel{(z^2 - xy)\bar{k}}$ ~~is~~ $(4x + cy + 2z)\bar{k}$

(1)

A) given $\bar{A} = (x+2y+az)\bar{i} + (bx-3y-z)\bar{j} + (4x+cy+2z)\bar{k}$

$= A_1\bar{i} + A_2\bar{j} + A_3\bar{k}$ (say)

$$\text{curl}\,\bar{A} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}$$

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y+az & bx-3y-z & 4x+cy+2z \end{vmatrix}$$

$$= \bar{i}\left[\frac{\partial}{\partial y}(4x+cy+2z) - \frac{\partial}{\partial z}(bx-3y-z)\right] - \bar{j}\left[\frac{\partial}{\partial x}(4x+cy+2z) - \frac{\partial}{\partial z}(x+2y+az)\right] + \bar{k}\left[\frac{\partial}{\partial x}(bx-3y-z) - \frac{\partial}{\partial y}(x+2y+az)\right]$$

$$= \bar{i}[c-1] - \bar{j}[4-a] + \bar{k}[b-2]$$

$\text{curl}\,\bar{A} = \bar{0}$

$\bar{i}[c-1] - \bar{j}[4-a] + \bar{k}[b-2] = \bar{0}$

$c-1=0$, $a-4=0$, $b-2=0$

$\boxed{c=1}$ $\boxed{a=4}$ $\boxed{b=2}$

hence $\bar{A}$ is irrotational

Since $\bar{A}$ irrotational hence their exists as Scalar potential function ϕ such that $\bar{A} = \nabla\phi$

$$\bar{A} = \nabla\phi.$$

$(x+2y+4z)\bar{i}+(2x-3y-z)\bar{j}+(4x+y+2z)\bar{K} = \frac{\partial\phi}{\partial x}\bar{i}+\frac{\partial\phi}{\partial y}\bar{j}+\frac{\partial\phi}{\partial z}\bar{K}$

$\frac{\partial\phi}{\partial x} = x+2y+4z, \frac{\partial\phi}{\partial y} = 2x-3y-z, \frac{\partial\phi}{\partial z} = 4x+y+2z$

$\frac{\partial\phi}{\partial x} \xrightarrow[(y,z\text{ cont.})]{\text{I.O.B w.r.t } x} \frac{x^2}{2}+2xy+4xz+g_1(y,z)$

$\frac{\partial\phi}{\partial y} \xrightarrow[(x,z\text{ as cont.})]{\text{I.O.B w.r.t } y} 2xy-3\frac{y^2}{2}-yz+g_2(x,z)$

$\frac{\partial\phi}{\partial z} \xrightarrow[(y,x\text{ as cont.})]{\text{I.O.B w.r.t } z} 4xz+yz+z^2+g_3(y,x)$

$\therefore \phi = \frac{x^2}{2}+2xy+4xz-\frac{3y^2}{2}-yz+z^2$

15) P.T (i) div(curl $\bar{f}$) = 0 (ii) Curl (grad ϕ) = 0

A) i) Let $\bar{f} = f_1\bar{i}+f_2\bar{j}+f_3\bar{K}$

$$\text{curl}\,\bar{f} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{K} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$\text{curl}\,\bar{f} = \bar{i}\left(\frac{\partial f_3}{\partial y}-\frac{\partial f_2}{\partial z}\right)-\bar{j}\left(\frac{\partial f_3}{\partial x}-\frac{\partial f_1}{\partial z}\right)+\bar{K}\left(\frac{\partial f_2}{\partial x}-\frac{\partial f_1}{\partial y}\right) \rightarrow (1)$$

we have $\text{div}(\bar{f}) = \frac{\partial f_1}{\partial x}+\frac{\partial f_2}{\partial y}+\frac{\partial f_3}{\partial z}$

eq-① comparer $\bar{g} = g_1\bar{i} + g_2\bar{j} + g_3\bar{k}$

$$\therefore \operatorname{div}(\operatorname{curl}\bar{F}) = \frac{\partial}{\partial x}\left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right) - \frac{\partial}{\partial y}\left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z}\right) + \frac{\partial}{\partial z}\left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)$$

$$= \cancel{\frac{\partial^2 f_3}{\partial x \partial y}} - \cancel{\frac{\partial^2 f_2}{\partial x \partial z}} - \cancel{\frac{\partial^2 f_3}{\partial x \partial y}} + \cancel{\frac{\partial^2 f_1}{\partial y \partial z}} + \cancel{\frac{\partial^2 f_2}{\partial z \partial x}} - \frac{\partial^2 f_1}{\partial z \partial y}$$

$$= \bar{0}$$

$$\therefore \operatorname{div}(\operatorname{curl}\bar{F}) = 0$$

ii) let ϕ be a scalar function

$$\therefore \operatorname{grad}\phi = \frac{\partial \phi}{\partial x}\bar{i} + \frac{\partial \phi}{\partial y}\bar{j} + \frac{\partial \phi}{\partial z}\bar{k}$$

$$\operatorname{Curl}(\operatorname{grad}\phi) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= \bar{i}\left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial y \partial z}\right) - \bar{j}\left(\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial z \partial x}\right) + \bar{k}\left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x}\right)$$

$$= 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$$= \bar{0}$$

$$\therefore \operatorname{Curl}(\operatorname{grad}\phi) = \bar{0}$$

16) show that $\operatorname{div}(\operatorname{grad} r^n) = \nabla^2(r^n) = n(n+1)r^{n-2}$ where $\bar{r}$ $r = |\bar{r}|$.

A) Let $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$$r = |\bar{r}| = \sqrt{x^2+y^2+z^2}$$

$$r^2 = x^2+y^2+z^2 \rightarrow ①$$

diff. eq ① w.r.t x

$$2r\frac{\partial r}{\partial x} = 2x$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

$|||^{ly}$ $\frac{\partial r}{\partial y} = \frac{y}{r}$, $\frac{\partial r}{\partial z} = \frac{z}{r}$

$$\nabla f(r) = \bar{i}\frac{\partial}{\partial x}f(r) + \bar{j}\frac{\partial}{\partial y}f(r) + \bar{k}\frac{\partial}{\partial z}f(r)$$

$$= \bar{i}f'(r)\frac{\partial f}{\partial x} + \bar{j}f'(r)\frac{\partial f}{\partial y} + \bar{k}f'(r)\frac{\partial f}{\partial z}$$

$$= \bar{i}f'(r)\frac{x}{r} + \bar{j}f'(r)\frac{y}{r} + \bar{k}f'(r)\frac{z}{r}$$

let

$$\bar{f} = \nabla f(r) = \bar{i}f'(r)\frac{x}{r} + \bar{j}f'(r)\frac{y}{r} + \bar{k}f'(r)\frac{z}{r}$$

$$= f_1\bar{i} + f_2\bar{j} + f_3\bar{k} \quad \text{(say)}$$

$$\therefore \operatorname{div}\bar{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$\operatorname{div}\operatorname{grad}\bar{f} = \frac{\partial}{\partial x}\left[\frac{x}{r}f'(r)\right] + \frac{\partial}{\partial y}\left[\frac{y}{r}f'(r)\right] + \frac{\partial}{\partial z}\left[\frac{z}{r}f'(r)\right]$$

consider

$$\frac{\partial}{\partial x}\left[\underset{u}{\frac{x}{r}}\underset{v}{f'(r)}\right] = \frac{x}{r}f''(r)\frac{\partial r}{\partial x} + f'(r)\left\{\frac{\partial}{\partial x}\left(\frac{x}{r}\right)\right.$$

$$= \frac{x}{r}f''(r)\frac{x}{r} + f'(r)\left[\frac{r\cdot 1 - x\frac{\partial r}{\partial x}}{r^2}\right]$$

$$= \frac{x^2}{r^2} f''(r) + f'(r)\left[\frac{r - x\frac{x}{r}}{r^2}\right]$$

$$= \frac{x^2}{r^2} f''(r) + f'(r)\left[\frac{r^2-x^2}{r^3}\right]$$

$$\frac{\partial}{\partial x}\left[\frac{x}{r} f'(r)\right] = \frac{x^2}{r^2} f''(r) + f'(r)\left[\frac{r^2-x^2}{r^3}\right]$$

|||ly

$$\frac{\partial}{\partial y}\left[\frac{y}{r} f'(r)\right] = \frac{y^2}{r^2} f''(r) + f'(r)\left[\frac{r^2-y^2}{r^3}\right]$$

$$\frac{\partial}{\partial z}\left[\frac{z}{r} f'(r)\right] = \frac{z^2}{r^2} f''(r) + f'(r)\left(\frac{r^2-z^2}{r^3}\right)$$

$$\therefore \nabla^2 f(r) = \frac{f''(r)}{r^2}(x^2+y^2+z^2) + \frac{f'(r)}{r^3}\left(r^2-x^2+r^2-y^2+r^2-z^2\right)$$

$$= f''(r) + \frac{f'(r)}{r^3}\left[3r^2-(x^2+y^2+z^2)\right]$$

$$= f''(r) + \frac{f'(r)}{r^3}\left[3r^2-r^2\right]$$

$$= f''(r) + \frac{f'(r)}{r^3}\left[2r^2\right]$$

$$\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r) \rightarrow (2)$$

given $f(r) = r^n$

$$f'(r) = n r^{n-1}$$

$$f''(r) = n(n-1) r^{n-2}$$

Now sub in eq (2) in $f'(r)$ & $f''(r)$ & $f(r)$

$$\nabla^2 r^n = n(n-1) r^{n-2} + \frac{2}{r} n r^{n-1}$$

$$= n\left[(n-1) r^{n-2} + 2\right] r^{n-2}$$

$$\nabla^2 r^n = n(n+1) r^{n-2}$$

17) show that $\nabla^2[f(r)] = f''(r) + \frac{2}{r} f'(r)$ where $r = |\bar{r}|$

A) Let $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$$r = |\bar{r}| = \sqrt{x^2+y^2+z^2}$$

$$r^2 = x^2 + y^2 + z^2 \rightarrow (1)$$

diff. eq (1) w.r.t 'x'

$$2r \frac{\partial r}{\partial x} = 2x$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

$|||^{ly}$

$$\frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\nabla f(r) = \bar{i}\frac{\partial}{\partial x} f(r) + \bar{j}\frac{\partial}{\partial y} f(r) + \bar{k}\frac{\partial}{\partial z} f(r)$$

$$= \bar{i} f'(r) \frac{\partial r}{\partial x} + \bar{j} f'(r) \frac{\partial r}{\partial y} + \bar{k} f'(r) \frac{\partial r}{\partial z}$$

$$= \bar{i} f'(r) \frac{x}{r} + \bar{j} f'(r) \frac{y}{r} + \bar{k} f'(r) \frac{z}{r}$$

Let

$$\bar{f} = \nabla f(r) = \bar{i} f'(r) \frac{x}{r} + \bar{j} f'(r) \frac{y}{r} + \bar{k} f'(r) \frac{z}{r}$$

$$= f_1 \bar{i} + f_2 \bar{j} + f_3 \bar{k} \quad \text{(say)}$$

$$\operatorname{div} \bar{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$\operatorname{div} \operatorname{grad} \bar{f} = \frac{\partial}{\partial x}\left[\frac{x}{r} f'(r)\right] + \frac{\partial}{\partial y}\left[\frac{y}{r} f'(r)\right] + \frac{\partial}{\partial z}\left[\frac{z}{r} f'(r)\right]$$

Consider

$$\frac{\partial}{\partial x}\left[\frac{x}{r} f'(r)\right] = \frac{x}{r} f''(r) \frac{\partial r}{\partial x} + f'(r) \frac{\partial}{\partial x}\left(\frac{x}{r}\right)$$

$$= \frac{x}{r} f''(r) \frac{x}{r} + f'(r)\left[\frac{r \cdot 1 - x \frac{\partial r}{\partial x}}{r^2}\right]$$

$$= \frac{x^2}{r^2} f''(r) + f'(r) \left[\frac{r - x\frac{x}{r}}{r^2}\right]$$

$$= \frac{x^2}{r^2} f''(r) + f'(r) \left[\frac{r - \frac{x^2}{r}}{r^2}\right]$$

$$\frac{\partial}{\partial x}\left[\frac{x}{r} f'(r)\right] = \frac{x^2}{r^2} f''(r) + f'(r) \left(\frac{r^2 - x^2}{r^3}\right)$$

|||ly

$$\frac{\partial}{\partial y}\left[\frac{y}{r} f'(r)\right] = \frac{y^2}{r^2} f''(r) + f'(r) \left(\frac{r^2 - y^2}{r^3}\right)$$

$$\frac{\partial}{\partial z}\left[\frac{z}{r} f'(r)\right] = \frac{z^2}{r^2} f''(r) + f'(r) \left(\frac{r^2 - z^2}{r^3}\right)$$

$$\therefore \nabla^2 f(r) = \frac{f''(r)}{r^2} (x^2 + y^2 + z^2) + \frac{f'(r)}{r^3} \left[r^2 - x^2 + r^2 - y^2 + r^2 - z^2\right]$$

$$= f''(r) + \frac{f'(r)}{r^3} \left[3r^2 - (x^2 + y^2 + z^2)\right]$$

$$= f''(r) + \frac{f'(r)}{r^3} \left[3r^2 - r^2\right]$$

$$= f''(r) + \frac{f'(r)}{r^3} \left[2\cancel{r^2}\right]$$

$$\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$$

(1)

Assignment-5

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1) If $\bar{F} = 3xy\bar{i} - y^2\bar{j}$, evaluate $\int \bar{F}\cdot d\bar{r}$ where c is the curve $y=2x^2$ in xy-plane from (0,0) to (1,2)

A) Given $\bar{F} = 3xy\bar{i} - y^2\bar{j}$

$\bar{F}\cdot d\bar{r} = (3xy\bar{i} - y^2\bar{j})(dx\bar{i} + dy\bar{j})$

$\therefore \bar{F}\cdot d\bar{r} = 3xy\,dx - y^2\,dy$

Given that c is the curve

$y=2x^2$ in xy-plane from (0,0) to (1,2)

$dy = 4x\,dx$

$\bar{F}\cdot d\bar{r} = 3x(2x^2)dx - (2x^2)^2(4x\,dx)$

$= 6x^2\,dx - \cancel{8x^3dx}\,(4x^4)(4x\,dx)$

$\cancel{= (6x^2 - 8x^3)}$

$= 6x^3dx - 16x^5\,dx$

$\bar{F}\cdot d\bar{r} = (6x^3 - 16x^5)\,dx$

along this curve x varies from 0 to 1

$$\int_c \bar{F}\cdot d\bar{r} = \int_{x=0}^{1} (6x^3 - 16x^5)\,dx$$

$$= 6\left(\frac{x^4}{4}\right)_{x=0}^{1} - 16\left(\frac{x^6}{6}\right)_{x=0}^{1}$$

$$= 6\left(\frac{1}{4} - 0\right) - 16\left(\frac{1}{6} - 0\right)$$

$$= \frac{3}{2} - \frac{8}{3}$$

$$= \frac{9-16}{6}$$

$$\int_C \bar{F} \cdot d\bar{r} = -7/6$$

2) Find the work done in moving a particle in the force field $\bar{F} = 3x^2\bar{i} + (2xz - y)\bar{j} + z\bar{k}$ along the straight line from $(0,0,0)$ to $(2,1,3)$

A) given that $\bar{F} = 3x^2\bar{i} + (2xz - y)\bar{j} + z\bar{k}$

$$\bar{F} \cdot d\bar{r} = (3x^2\bar{i} + (2xz-y)\bar{j} + z\bar{k})(dx\bar{i} + dy\bar{j} + dz\bar{k})$$

$$= 3x^2\,dx + (2xz - y)\,dy + z\,dz$$

Let $O = (0,0,0)$

$A = (2,1,3)$

given that C is the ~~curve~~ joining the straight lines ~~from OA~~ $O(0,0,0)$ & $A(2,1,3)$

$$\cancel{\boxed{\int_C \bar{F} \cdot d\bar{r} = \int}}$$

$$\frac{x_2 - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

$$\frac{x-0}{2-0} = \frac{y-0}{1-0} = \frac{z-0}{3-0}$$

$$\frac{x}{2} = \frac{y}{1} = \frac{z}{3} = t$$

$x = 2t, \quad y = t, \quad z = 3t$

(2)

$x = 2t \quad y = t \quad z = 3t$

$dx = 2dt \quad dy = dt \quad dz = 3dt$

$(x, y, z) = (0,0,0)$ to $(2,1,3)$

$(2t, t, 3t) = (0,0,0)$ to $(2,1,3)$

$2t = 0$ & $2t = 2$

$t = 0, 1,$

<u>Now</u>

$$W = \int_C \bar{F} \cdot d\bar{r} = \int 3x^2 dx + (2xz - y)\,dy + z\,dz$$

$$= \int_{t=0}^{1} 3(2t)^2 (2dt) + (2(2t)(3t) - t)(dt) + (3t)(3dt)$$

$$= \int_{t=0}^{1} \cancel{(6t^2)}\, 3(4t^2)(2dt) + (\cancel{2}(6t^2) - t)(dt) + 9t\,dt)$$

$$= \int_{t=0}^{1} 24t^2 dt + (12t^2 - t)dt + 9t\,dt$$

$$= \int_{t=0} (24t^2 + 12t^2 - t + 9t)\,dt$$

$$= \int_{t=0}^{1} (36t^2 + 8t)\,dt$$

$$= \overset{12}{\cancel{36}} \left[\frac{t^3}{\cancel{3}}\right]_{t=0}^{1} + \overset{4}{\cancel{8}} \left[\frac{t^2}{\cancel{2}}\right]_{t=0}^{1}$$

$= 12(1-0) + 4(1-0)$

$W = 16$ units

3) Find the work done by the force field $\overline{F} = z\overline{i} + x\overline{j} + y\overline{k}$ when it moves a particle along the arc of the curve $\overline{r} = \cos t\,\overline{i} + \sin t\,\overline{j} - t\overline{k}$ from $t=0$ to $t=2\pi$

A) given curve $\overline{F} = z\overline{i} + x\overline{j} + y\overline{k}$

$$\overline{r} = \cos t\,\overline{i} + \sin t\,\overline{j} - t\overline{k}$$

$$x = \cos t,\quad y = \sin t,\quad z = -t$$

$$d\overline{r} = -\sin t\,\overline{i}\,dt + \cos t\,\overline{j}\,dt - \overline{k}\,dt$$

$$\overline{F} = (-t)\overline{i} + (\cos t)\overline{j} + (\sin t)\overline{k}$$

$$\overline{F}\cdot d\overline{r} = (-t\,\overline{i} + \cos t\,\overline{j} + \sin t\,\overline{k})(-\sin t\,\overline{i}\,dt + \cos t\,\overline{j}\,dt - \overline{k}\,dt)$$

$$\overline{F}\cdot d\overline{r} = t\sin t\,dt + \cos^2 t\,dt - \sin t\,dt$$

$$\boxed{\cancel{\int_0^c \overline{F}\cdot d\overline{r} = \int_0^{2\pi} [t\sin t\,dt + \cos^2 t\,dt - \sin t\,dt] = \int_0^{2\pi} (t\sin t + \cos^2 t - \sin t)\,dt}}$$

$$\overline{F}\cdot d\overline{r} = t\sin t\,dt + \left(\frac{1+\cos 2t}{2}\right)dt - \sin t\,dt$$

$$\overline{F}\cdot d\overline{r} = t\sin t\,dt + \frac{1}{2}dt + \frac{1}{2}\cos 2t\,dt - \sin t\,dt$$

$$\int_0^c \overline{F}\cdot d\overline{r} = \int_0^{2\pi} \left[t\sin t\,dt + \frac{1}{2}dt + \frac{1}{2}\cos 2t\,dt - \sin t\,dt\right]$$

$$= \left[-t\cos t - \int -\cos t\right]_0^{2\pi} + \frac{1}{2}(t)_0^{2\pi} + \frac{1}{2}\left[\frac{\sin 2t}{2}\right]_0^{2\pi} - [-\cos t]_0^{2\pi}$$

$$= [-t\cos t + \sin t]_0^{2\pi} + \frac{1}{2}(2\pi) + \frac{1}{4}(\sin 2t)_0^{2\pi} + [\cos t]_0^{2\pi}$$

$$= \cancel{[-t\cos t + \sin t]_0^{2\pi}}$$

③

$= [-2\pi \cos(2\pi) + \sin(2\pi) - t\cos(0)] + \pi + \frac{1}{4}[\sin 4\pi] +$
$\cos 2\pi - \cos 0$

$= -2\pi(1) + 0 - \cancel{t} + \pi + \frac{1}{4}(0) + 1 - \cancel{1}$

$= -2\pi + \pi$

$\int_0^c \bar{F}\cdot d\bar{r} = -\pi$

4) Evaluate $\int_S \bar{F}\cdot\bar{n}\, ds$ where $\bar{F} = 12x^2y\bar{i} - 3yz\bar{j} + 2z\bar{k}$ and s is the portion of the plane $x+y+z=1$ included in the 1st octant.

A) given that $\bar{F} = 12x^2y\bar{i} - 3yz\bar{j} + 2z\bar{k}$
Given that S is the part of the surface of the plane $x+y+z=1 \rightarrow (1)$
located in the 1st octant.

$\phi = x+y+z-1$

$\therefore \nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$

$\nabla\phi = \bar{i}(1) + \bar{j}(1) + \bar{k}(1)$

$\nabla\phi = \bar{i} + \bar{j} + \bar{k}$

$\therefore$ unit normal vector

$\bar{n} = \frac{\nabla\phi}{|\nabla\phi|}$

$= \frac{\bar{i}+\bar{j}+\bar{k}}{\sqrt{1+1+1}}$

$\bar{n} = \frac{1}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})$

Let 'R' be the projection of 'S' on xy plane

$$\therefore \int_S \bar{F}\cdot\bar{n}\,ds = \iint_R \frac{\bar{F}\cdot\bar{n}}{|\bar{n}\cdot\bar{k}|}\,dx\,dy$$

Now

$$\bar{F}\cdot\bar{n} = (12x^2y\bar{i} - 3yz\bar{j} + 2z\bar{k})\cdot\left(\frac{1}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})\right)$$

$$= \frac{1}{\sqrt{3}}(12x^2y - 3yz + 2z)$$

from ①

$$z = 1 - x - y$$

$$\bar{F}\cdot\bar{n} = \frac{1}{\sqrt{3}}(12x^2y - 3y(1-x-y) + 2(1-x-y))$$

$$= \frac{1}{\sqrt{3}}(12x^2y - 3y + 3xy + 3y^2 + 2 - 2x - 2y)$$

$$\bar{F}\cdot\bar{n} = \frac{1}{\sqrt{3}}(12x^2y + 3y^2 - 5y + 3xy - 2x + 2)$$

$$\bar{n}\cdot\bar{k} = \frac{1}{\sqrt{3}}(\bar{i}+\bar{j}+\bar{k})\cdot\bar{k}$$

$$|\bar{n}\cdot\bar{k}| = \frac{1}{\sqrt{3}}$$

$\int_S$ ~~F.n~~ ds The projection 'R' of 'S' in xy-plane is given by $x + y + \cancel{0} = 1$ $(\because z = 0)$

In the region 'R' y varries from 0 1 - x &

x varries from 0 to 1

(4)

$$\therefore \int_S \bar{F}\cdot\bar{n}\, dS = \iint_R \frac{\bar{F}\cdot\bar{n}}{|\bar{n}\cdot\bar{k}|}\, dx\, dy$$

$$= \int_{x=0}^{1}\int_{y=0}^{1-x} (12x^2y + 3y^2 - 5y + 3xy - 2x + 2)\, dy\, dx$$

$$= \int_{x=0}^{1} \left[\cancel{12}^{6}x^2 \frac{y^2}{\cancel{2}} + \cancel{3}\frac{y^3}{\cancel{3}} - 5\frac{y^2}{2} + 3x\frac{y^2}{2} - 2xy + 2y \right]_{y=0}^{1-x} dx$$

$$= \int_{x=0}^{1} \left[6x^2y^2 + y^3 - \frac{5}{2}y^2 + \frac{3}{2}xy^2 - 2xy + 2y \right]_{y=0}^{1-x} dx$$

$$\cancel{= \int_{x=0}^{1} 6x^2(1-x)^2 + (1-x)^3 - \frac{5}{2}(1-x)^2 + \frac{3}{2}x(}$$

$$= \int_{x=0}^{1} 6x^2(1-x)^2 + (1-x)^3 - \frac{5}{2}(1-x)^2 + \frac{3}{2}x(1-x)^2 - 2x(1-x) + 2(1-x) \Big]_{y=0}^{1-x} dx$$

$$= \int_{x=0}^{1} 6x^2(1+x^2-2x) + (1-3x+3x^2-x^3) - \frac{5}{2}(1+x^2-2x) + \frac{3}{2}x(1+x^2-2x) - 2x + 2x^2 + 2 - 2x \Big]_{y=0}^{1-x} dx$$

$$= \int_{x=0}^{1} 6x^2 + 6x^4 - 12x^3 + 1 - 3x + 3x^2 - x^3 - \frac{5}{2} - \frac{5}{2}x^2 + 5x + \frac{3}{2}x + \frac{3}{2}x^3 - 3x^2 - 4x + 2x^2 + 2 \Big]_{y=0}^{1-x} dx$$

$$= \int_{x=0}^{1} 6x^2 + 6x^4 - 12x^3 + 1 - 3x + \cancel{3x^2} - x^3 - \frac{5}{2} - \frac{5}{2}x^2 + 5x + \frac{3}{2}x$$
$$+ \frac{3}{2}x^3 - \cancel{3x^2} - 4x + 2x^2 + 2 \Big]_{y=0}^{1-x} dx$$

$$= \int_{x=0}^{1} \left[6x^4 - 13x^3 + \frac{3}{2}x^3 + 8x^2 - \frac{5}{2}x^2 - 2x + \frac{3}{2}x + 3 - \frac{5}{2}\right] dx$$

$$= \Big[6\frac{x^5}{5} - 13\frac{x^4}{4} + \frac{3}{2}\frac{x^4}{4} + 8\frac{x^3}{3} - \frac{5}{2}\frac{x^3}{3} - \cancel{2}\frac{x^2}{\cancel{2}} + \frac{3}{2}\frac{x^2}{2}$$
$$+ 3x - \frac{5}{2}x\Big]_{x=0}^{1} dx$$

$$= \left[\frac{6}{5} - \frac{13}{4} + \frac{3}{8} + \frac{8}{3} - \frac{5}{6} - 1 + \frac{3}{4} + 3 - \frac{5}{2} - (0)\right]$$

$$= \left[\frac{3}{8} - \frac{5}{6} + \frac{6}{5} - \frac{10}{4} + \frac{8}{3} - \frac{5}{2} + 2\right]$$

$$= \frac{45 - 100 + 144 - 300 + 320 - 300 + 240}{120}$$

$$= \cancel{\frac{-700 + 689}{120}} = \frac{749 - 700}{120}$$

$$\cancel{= -11/120} \quad = 49/120$$

$$\int_S F \cdot n \, ds = 49/120$$

⑤

5) evaluate $\int_S \bar{F}\cdot\bar{n}\, ds$ where $\bar{F} = z\bar{i} + x\bar{j} - 3y^2 z\bar{k}$ & S is the surface $x^2+y^2=16$ included in the first octant b/w $z=0$ & $z=5$.

A) given that $\bar{F} = z\bar{i} + x\bar{j} - 3y^2 z\bar{k}$

and 'S' is the surface $x^2+y^2=16$ included in the first octant b/w $z=0$ & $z=5$

Let $\phi = x^2+y^2-16$

$$\nabla\phi = \bar{i}\frac{\partial\phi}{\partial x} + \bar{j}\frac{\partial\phi}{\partial y} + \bar{k}\frac{\partial\phi}{\partial z}$$

$$= \bar{i}(2x) + \bar{j}(2y) + \bar{k}(0)$$

$$\nabla\phi = 2x\bar{i} + 2y\bar{j}$$

∴ unit ~~vec~~ normal vector

$$\bar{n} = \frac{\nabla\phi}{|\nabla\phi|}$$

$$= \frac{2x\bar{i}+2y\bar{j}}{\sqrt{4x^2+4y^2}}$$

$$= \frac{2(x\bar{i}+y\bar{j})}{2\sqrt{x^2+y^2}}$$

$$= \frac{(x\bar{i}+y\bar{j})}{\sqrt{16}} \qquad (\because x^2+y^2=16)$$

$$\bar{n} = \frac{1}{4}(x\bar{i}+y\bar{j})$$

Let 'R' be the projection of 'S' on yz plane

$$\therefore \int_S \bar{F}\cdot\bar{n}\,ds = \iint_R \frac{\bar{F}\cdot\bar{n}}{|\bar{n}\cdot\bar{i}|}\,dy\,dz$$

<u>Now</u>

$$\bar{F}\cdot\bar{n} = (z\bar{i}+x\bar{j}-3y^2z\bar{k})\cdot\frac{1}{4}(x\bar{i}+y\bar{j})$$

$$= \frac{1}{4}(xz+xy)$$

$$\bar{F}\cdot\bar{n} = \frac{x}{4}(z+y)$$

$$\bar{n}\cdot\bar{i} = \frac{1}{4}(x\bar{i}+y\bar{j})\cdot\bar{i}$$

$$|\bar{n}\cdot\bar{i}| = \frac{1}{4}x$$

The projection 'R' & 'S' on yz plane is $y^2=16$ ($\because x=0$)

$$\int_S \bar{F}\cdot\bar{n}\,ds = \int_{y=0}^{4}\int_{z=0}^{5} \frac{\cancel{\frac{x}{4}}(z+y)}{\cancel{\frac{x}{4}}}\,dz\,dy$$

$$= \int_{y=0}^{4}\left(\frac{z^2}{2}+zy\right)_{z=0}^{5} dy$$

$$= \int_{y=0}^{4}\left(\frac{25}{2}+5y\right)dy$$

$$= \left[\frac{25}{2}y+5\frac{y^2}{2}\right]_{y=0}^{4}$$

$$= \frac{25(4)}{2}+\frac{5(4)}{2}$$

$$= \frac{100}{2}+\frac{80}{2}$$

$$\int_S \bar{F}\cdot\bar{n}\,ds = 90$$

(c)

6) Evaluate $\int_V \bar{F}\cdot dv$, given that $\bar{F} = 2xz\bar{i} - x\bar{j} + y^2\bar{k}$ & V is the region bounded by the surfaces $x=0, x=2; y=0, y=6$ & $z=x^2, z=4$

A) given $\bar{F} = 2xz\bar{i} - x\bar{j} + y^2\bar{k}$

In the region 'z' varries from x^2 to 4, 'y' varries from 0 to 6 & 'x' varries from 0 to 2.

$$\boxed{\cancel{\int_V \operatorname{div}\bar{F}\,dV = \int_{x=0}^{2}\int_{y=0}^{6}\int_{z=x^2}^{4} (2xz\bar{i} - x\bar{j} + y^2\bar{k})\,dz\,dy\,dx}}$$

$$\int_{x=0}^{2}\int_{y=0}^{6}\int_{z=x^2}^{4} (2xz\bar{i} - x\bar{j} + y^2\bar{k})\,dz\,dy\,dx = \int_V \bar{F}\cdot dV$$

$$\int_V \bar{F}\cdot d\bar{v} = \int_{x=0}^{2}\int_{y=0}^{6}\left[2x\frac{z^2}{2}\bar{i} - xz\bar{j} + y^2 z\bar{k}\right]_{z=x^2}^{4} dy\,dx$$

$$= \int_{x=0}^{2}\int_{y=0}^{6}\left[xz^2\bar{i} - xz\bar{j} + y^2 z\bar{k}\right]_{z=x^2}^{4} dy\,dx$$

$$= \int_{x=0}^{2}\int_{y=0}^{6}\left[x(4)^2\bar{i} - x(4)\bar{j} + y^2(4)\bar{k} - x(x^4)\bar{i} + x x^2\bar{j} - y^2x^2\bar{k}\right] dy\,dx$$

$$= \int_{x=0}^{2}\int_{y=0}^{6}\left[16x\bar{i} - 4x\bar{j} + 4y^2\bar{k} - x^5\bar{i} + x^3\bar{j} - y^2x^2\bar{k}\right] dy\,dx$$

$$= \int_{x=0}^{2}\int_{y=0}^{6}\left[[16x - x^5]\bar{i} + (x^3-4x)\bar{j} + (4y^2 - y^2x^2)\bar{k}\right] dy\, dx$$

$$= \int_{x=0}^{2} \left[[16xy - x^5y]\bar{i} + (x^3-4x)y\bar{j} + \left(4\frac{y^3}{3} - \frac{y^3}{3}x^2\right)\bar{k}\right]_{y=0}^{6} dx$$

$$= \int_{x=0}^{2}\left[(16x-x^5)(6)\bar{i} + (x^3-4x)(6)\bar{j} + \left(\frac{4}{3}-\frac{x^2}{3}\right)(6)^3\bar{k} - 0\right] dx$$

$$= \int_{x=0}^{2}\left[(96x-6x^5)\bar{i} + (6x^3-24x)\bar{j} + \left(\frac{4}{\cancel{3}}(\overset{72}{\cancel{216}}) - \frac{x^2}{\cancel{3}}(\overset{72}{\cancel{216}})\right)\bar{k}\right] dx$$

$$= \int_{x=0}^{2}\left[(96x-6x^5)\bar{i} + (6x^3-24x)\bar{j} + (288-72x^2)\bar{k}\right] dx$$

$$= \left[\left(96\frac{x^2}{2} - 6\frac{x^6}{6}\right)\bar{i} + \left(6\frac{x^4}{4} - 24\frac{x^2}{2}\right)\bar{j} + \left(288x - 72\frac{x^3}{3}\right)\bar{k}\right]_{x=0}^{2}$$

$$= \left[(48x^2 - x^6)\bar{i} + \left(\frac{3}{2}x^4 - 12x^2\right)\bar{j} + (288x - 24x^3)\bar{k}\right]_{x=0}^{6}$$

$$= \left(48(6)^2 - (6)^6\right)\bar{i} + \left(\frac{3}{2}(6)^4 - 12(6)^2\right)\bar{j} + \left(288(6) - 24(6)^3\right)\bar{k}$$

$$= \left(48(36) - 46656\right)\bar{i} + \left(\frac{3}{2}(1296) - 12(36)\right)\bar{j} + \left(1728 - 24(216)\right)\bar{k}$$

$$= 1728 -$$

$$= \left(48(2)^2 - (2)^6\right)\bar{i} + \left(\frac{3}{2}(2)^4 - 12(2)^2\right)\bar{j} + \left(288(2) - 24(2)^3\right)\bar{k}$$

$$= 128\bar{i} - 24\bar{j} + 384\bar{k}$$

$$\int_V \bar{F}\cdot dV = 128\bar{i} - 24\bar{j} + 384\bar{k}$$

7

7) verify green's theorem for $\int_c [(xy+y^2)dx + x^2dy]$, where c is bounded by $y=x$ & $y=x^2$.

A) By green's theorem we have

$$\oint_c Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

here. $M = xy+y^2$, $N = x^2$

Now $\frac{\partial M}{\partial y} = x+2y$ $\quad \frac{\partial N}{\partial x} = 2x$

Now $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x - x - 2y$

$= x - 2y$

Given that c is the boundary of the region bounded by the parameters

$y=x,\ y=x^2$

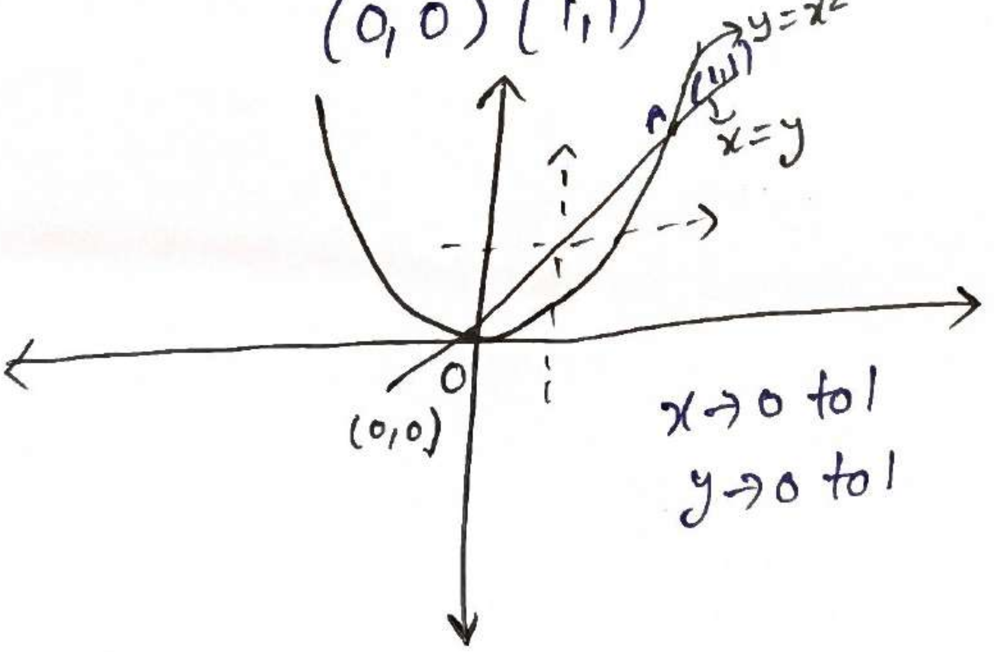
$x = x^2$

$x^2 - x = 0$

$x(x-1) = 0$

$x=0, x=1$ & $y=0, y=1$

$(0,0)\ (1,1)$



$x \to 0$ to 1 $\qquad x \to y$ to $\sqrt{y}$

$y \to 0$ to 1 $\qquad y \to x$ to x^2

To evaluate the line integral

i) along OA we have

$$y = x^2$$
$$dy = 2x\,dx$$

x varries from 0 to 1

$$\therefore M\,dx + N\,dy = (xy + y^2)\,dx + x^2\,dy$$
$$= (x x^2 + (x^2)^2)\,dx + x^2(2x\,dx)$$
$$= (x^3 + x^4)\,dx + 2x^3\,dx$$
$$= (x^3 + x^4 + 2x^3)\,dx$$
$$= (3x^3 + x^4)\,dx$$

$$\therefore \int_{OA} M\,dx + N\,dy = \int_0^1 (3x^3 + x^4)\,dx$$
$$= \left[3\frac{x^4}{4} + \frac{x^5}{5}\right]_0^1$$
$$= \frac{3(1)}{4} + \frac{1}{5}$$
$$= \frac{15+4}{20}$$
$$\int_{OA} M\,dx + N\,dy = \frac{19}{20}$$

ii) along AO we have

$$x = y$$
$$dx = dy$$

y varries from 1 to 0

(8)

$$Mdx + Ndy = (xy + y^2)dx + x^2dy$$
$$= (x(x) + x^2)dx + x^2dx$$
$$= (2x^2 + x^2)\,dx$$
$$= 3x^2dx$$

$$\therefore \int_{AO} Mdx + Ndy = \int_1^0 3x^2dx$$

$$= 3\frac{x^3}{3}\Big]_1^0$$

$$\int_{AO} Mdx + Ndy = -1$$

$$\therefore \oint_C Mdx + Ndy = \int_{OA} Mdx + Ndy + \int_{AO} Mdx + Ndy$$

$$= \frac{19}{20} - 1$$

$$= \frac{19-20}{20}$$

$$\boxed{\oint_C Mdx + Ndy = -1/20}$$

To evaluate the double integral

In the region x varries from 0 to 1 & y varries from x to x^2

$$= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

$$= \int_{x=0}^{1} \int_{y=x^2}^{x} (x-2y)\,dy\,dx$$

$$= \int_{x=0}^{1} \left[(xy - y^2)\right]_{y=x^2}^{x} dx$$

$$= \int_{x=0}^{1} (\cancel{x^2} - \cancel{x^2} - x^3 + x^4)\,dx$$

$$= \int_{x=0}^{1} (-x^3 + x^4)\,dx$$

$$= \left. \frac{-x^4}{4} + \frac{x^5}{5} \right]_{x=0}^{1}$$

$$= \frac{-1}{4} + \frac{1}{5}$$

$$= \frac{-5+4}{20}$$

$$= -1/20$$

$$\therefore \oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

hence green's theorem verified.

⑨

8) using Green's theorem ~~usin~~ evaluate $\int_c [(y-\sin x)dx + \cos x\, dy]$, where c is ~~bo~~ triangle enclosed by the lines $y=0$, $x=\pi/2$ & $y=\frac{2}{\pi}x$.

A) By green's theorem we have

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx dy$$

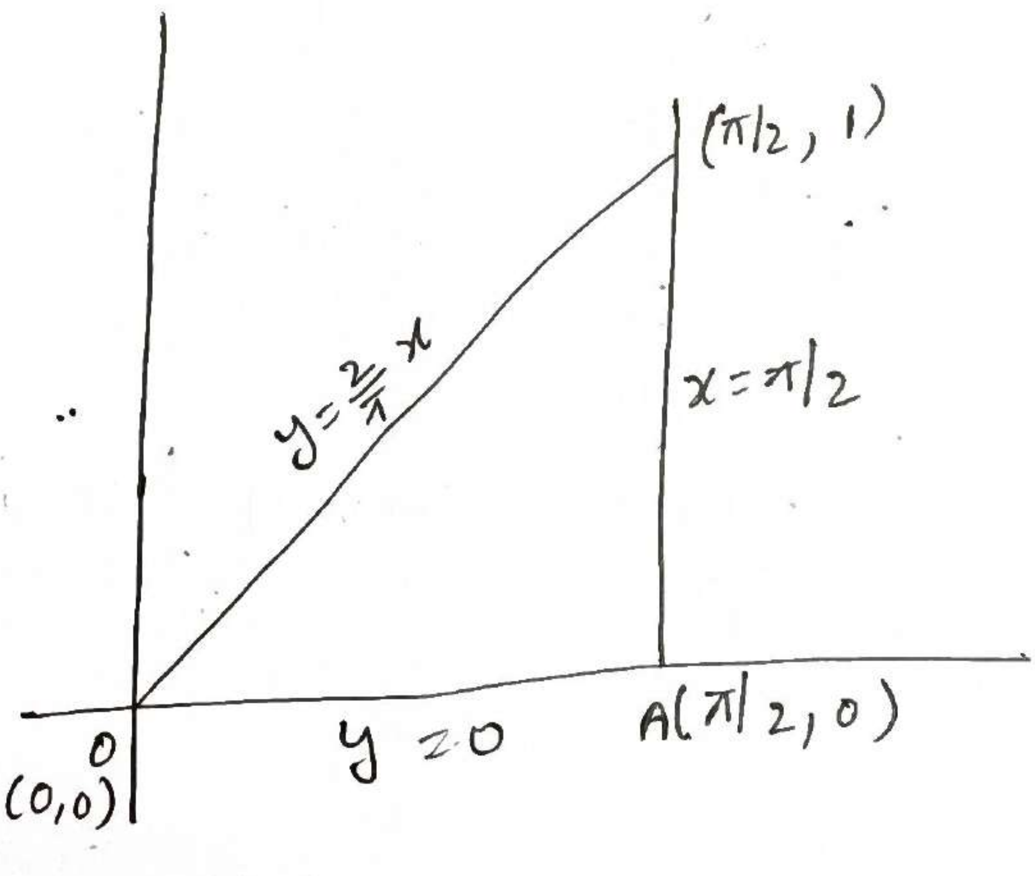
where $M = y - \sin x$ $\quad N = \cos x$

$$\frac{\partial M}{\partial y} = \cancel{\frac{y^2}{2}}\ 1 \qquad \frac{\partial N}{\partial x} = -\sin x$$

$$\therefore \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -\sin x - \cancel{\frac{y^2}{2}}\ 1$$

$$= -(1+\sin x)$$

given that 'c' is the ~~boundary of the region~~ triangle enclosed by lines $y=0$, $x=\pi/2$ & $y=\frac{2}{\pi}x$



$y = \frac{2}{\pi}x$

$y = \frac{2}{\pi} \times \frac{\pi}{2}$

$y = 1$

strip parallel to y-axis

$$y = 0 \text{ to } \frac{2}{\pi}x$$

$$x = 0 \text{ to } \pi/2$$

By green's theorem

$$\oint_C M\,dx + N\,dy = \iint_S \left[-(\sin x + 1)\right] dx\,dy$$

$$= -\int_{x=0}^{\pi/2} \int_{y=0}^{\frac{2}{\pi}x} (\sin x + 1)\, dy\, dx$$

$$= -\int_{x=0}^{\pi/2} \left[(\sin x + 1) y\right]_{y=0}^{\frac{2}{\pi}x} dx$$

$$= -\int_{x=0}^{\pi/2} \left[(\sin x + 1) \frac{2}{\pi} x\right] dx$$

$$= \frac{-2}{\pi} \int_{x=0}^{\pi/2} (x \sin x + x)\, dx$$

$$= \frac{-2}{\pi} \left[\int_{x=0}^{\pi/2} x \sin x\, dx + \int_{x=0}^{\pi/2} x\, dx\right]$$

Take

$$\int_{x=0}^{\pi/2} \overset{u}{x}\, \overset{v}{\sin x}\, dx = \left[-x\cos x + \sin x\right]_{x=0}^{\pi/2}$$

(10)

$$= \left[-\frac{\pi}{2}\cos\frac{\pi}{2} + \sin\frac{\pi}{2} - 0 + 0\right]$$

$$\int_{x=0}^{\pi/2} x\sin x\,dx = 1$$

Take

$$\int_{x=0}^{\pi/2} x\,dx = \left[\frac{x^2}{2}\right]_{x=0}^{\pi/2}$$

$$= \frac{\frac{\pi^2}{4}}{2}$$

$$\int_{x=0}^{\pi/2} x\,dx = \frac{\pi^2}{8}$$

Now

$$\oint_C M dx + N dy = \frac{-2}{\pi}\left[1 + \frac{\pi^2}{8}\right]$$

$$= \frac{-2}{\pi} - \frac{\pi}{4}$$

9) Verify stoke's theorem for $\bar{F} = (x^2+y^2)\bar{i} - 2xy\bar{j}$ taken around the rectangle bounded by the lines $x = \pm a$, $y = 0$ & $y = b$.

A) By stoke's theorem we have

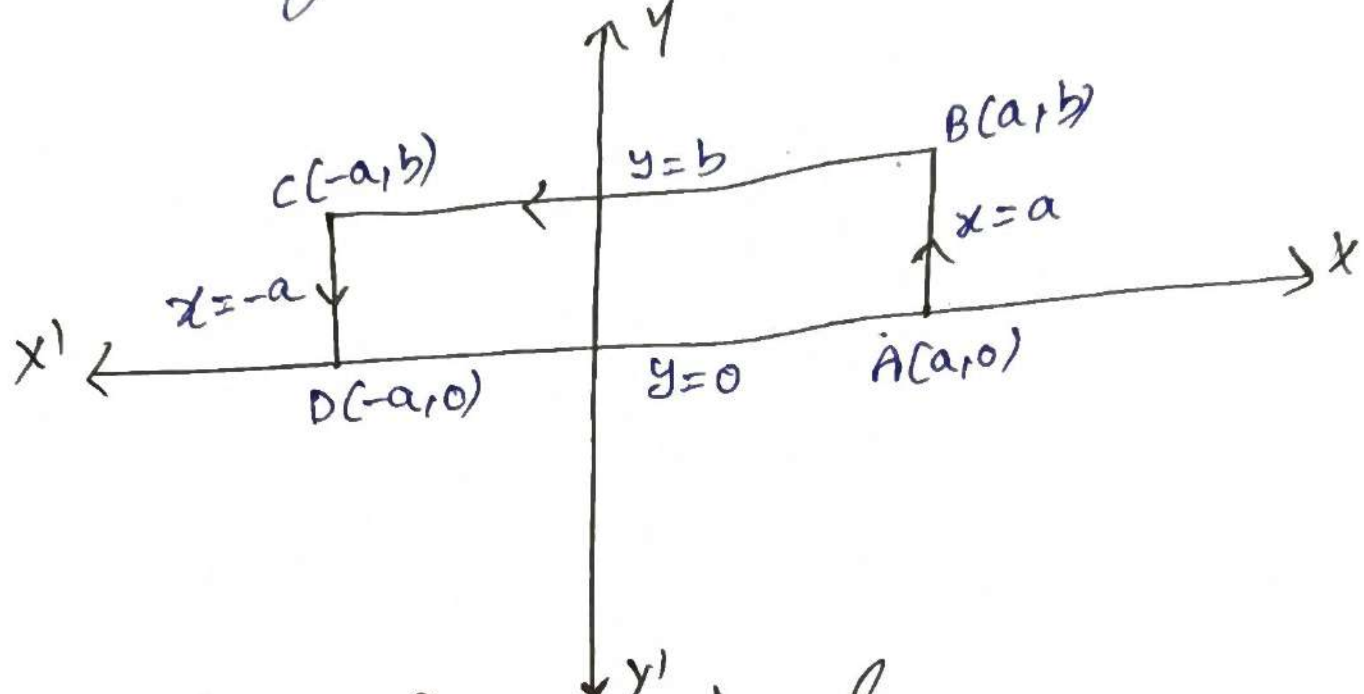
$$\oint_C \bar{F}\cdot d\bar{r} = \int_S \text{curl}\,\bar{F}\cdot\bar{n}\,ds$$

given that

$$\bar{F} = (x^2+y^2)\bar{i} - 2xy\bar{j}$$

$\bar{F}\cdot d\bar{r} = (x^2+y^2)\,dx - 2xy\,dy$

given that 'c' is the boundary of the rectangle bounded by the lines $x = \pm a$, $y=0$ & $y=b$



To evaluate the line integral

i) along the line AB we have

$$x = a$$
$$dx = 0$$

y varries from 0 to b.

$$\bar{F}\cdot d\bar{r} = -2ay\,dy$$

$$\therefore \int_{AB} \bar{F}\cdot d\bar{r} = \int_{y=0}^{b} -2ay\,dy$$

$$= -2a\left.\frac{y^2}{2}\right]_{y=0}^{b}$$

$$\int_{AB} \bar{F}\cdot d\bar{r} = -ab^2$$

ii) along the line BC we have

$$y = b$$
$$dy = 0$$

x varries from a to $-a$

$$\bar{F} \cdot d\bar{r} = (x^2 + b^2)\,dx$$

$$\int_{BC} \bar{F} \cdot d\bar{r} = \int_{x=a}^{-a} (x^2 + b^2)\,dx$$

$$= \left[\frac{x^3}{3} + b^2 x\right]_{x=a}^{-a}$$

$$\cancel{= \frac{a^3}{3} + b^2}$$

$$= -\frac{a^3}{3} - b^2 a - \frac{a^3}{3} - b^2 a$$

$$\int_{BC} \bar{F} \cdot d\bar{r} = -\frac{2a^3}{3} - 2ab^2$$

iii) along the line CD we have

$$x = -a$$
$$dx = 0$$

y varries from b to 0.

$$\bar{F} \cdot d\bar{r} = -2(-a)y\,dy$$

$$\bar{F} \cdot d\bar{r} = 2ay\,dy$$

$$\int_{CD} \bar{F} \cdot d\bar{r} = \int_{y=b}^{0} 2ay\,dy$$

$$= 2a \left[\frac{y^2}{2}\right]_{y=b}^{0}$$

$$= a(0)^2 - a(b)^2$$

$$\int_{CD} \bar{F} \cdot d\bar{r} = -ab^2$$

iv) along the DA we have

$$y = 0$$
$$dy = 0$$

x varries from $-a$ to a

$$\bar{F}\cdot d\bar{r} = x^2\,dx$$

$$\int_{DA} \bar{F}\cdot d\bar{r} = \int_{x=-a}^{a} x^2\,dx$$

$$= \left.\frac{x^3}{3}\right]_{x=-a}^{a}$$

$$= \frac{a^3}{3} + \frac{a^3}{3}$$

$$\int_{DA} F\cdot d\bar{r} = \frac{2}{3}a^3$$

$$\oint_C \bar{F}\cdot d\bar{r} = \int_{AB} \bar{F}\cdot d\bar{r} + \int_{BC} \bar{F}\cdot d\bar{r} + \int_{CD} \bar{F}\cdot d\bar{r} + \int_{DA} \bar{F}\cdot d\bar{r}$$

$$\oint_C \bar{F}\cdot d\bar{r} = -ab^2 - \cancel{\tfrac{2}{3}a^3} - 2ab^2 - ab^2 + \cancel{\tfrac{2}{3}a^3}$$

$$\oint_C F\cdot d\bar{r} = -4ab^2$$

To evaluate surface integral

$$\text{Curl}\,\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

(12)

$$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & -2xy & 0 \end{vmatrix}$$

$$= \bar{i}\left(0 - \frac{\partial}{\partial z}(-2xy)\right) - \bar{j}\left(\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(x^2+y^2)\right) + \bar{k}\left(\frac{\partial}{\partial x}(-2xy) - \frac{\partial}{\partial y}(x^2+y^2)\right)$$

$$= 0 - 0 + \bar{k}(-2y - 2y)$$

$\bar{n} = \bar{i} + \bar{j} + \bar{k}$

$\text{curl}\,\bar{F} = -4y\bar{k}$

$\text{curl}\,\bar{F} \cdot \bar{n} = -4y\bar{k} \cdot \bar{k}$ $\quad (\because n = \bar{k}$ for xy plane$)$

$\text{curl}\,\bar{F} \cdot \bar{n} = -4y$

$$\therefore \int_S \text{curl}\,\bar{F} \cdot \bar{n}\, ds = \int_{x=-a}^{a} \int_{y=0}^{b} -4y\, dy\, dx$$

$$= -4 \int_{x=-a}^{a} \left[\frac{y^2}{2}\right]_{y=0}^{b} dx$$

$$= -4 \int_{x=-a}^{a} \frac{b^2}{2}\, dx$$

$$= -2b^2 \left[x\right]_{-a}^{a}$$

$$= -2b^2[a+a]$$

$$\therefore \int_S \text{curl}\,\bar{F} \cdot \bar{n}\, ds = -4ab^2$$

$$\therefore \oint_C \bar{F}\cdot d\bar{r} = \int_S \text{curl}\,\bar{F}\cdot\bar{n}\,ds$$

hence stoke theorem is verified.

10) verify stoke's theorem for the vector field $\bar{F} = (2x-y)\bar{i} - yz^2\bar{j} - y^2z\bar{K}$ over the upper half of the sphere $x^2+y^2+z^2=1$ bounded by its projection on the xy-plane.

A) By stoke's theorem we have

$$\oint_C \bar{F}\cdot d\bar{r} = \int_S \text{curl}\,\bar{F}\cdot\bar{n}\,ds$$

Given that

$$\bar{F}\cdot d\bar{r} = (2x-y)dx - yz^2dy - y^2z\,dz \qquad (\because z=0, dz=0 \text{ in } xy\text{ plane})$$

$$\bar{F}\cdot d\bar{r} = (2x-y)\,dx$$

The boundary 'c' of 's' in xy-plane is given by $x^2+y^2=1$.

The parametric equation of circle $x^2+y^2=1$ is given by

$$x=\cos\theta,\ y=\sin\theta \quad (\because r=1)$$

$$dx=-\sin\theta\,d\theta,\ dy=\cos\theta\,d\theta$$

θ varries from 0 to 2π

$$\therefore \bar{F}\cdot d\bar{r} = [2\cos\theta-\sin\theta][-\sin\theta\,d\theta]$$

$$\bar{F}\cdot d\bar{r} = \left[-\sin 2\theta + \tfrac{1}{2}(1-\cos 2\theta)\right]d\theta$$

(13)

$$\therefore \oint_C \bar{F}\cdot d\bar{r} = \int_0^{2\pi} \left[-\sin 2\theta + \frac{1}{2}(1-\cos 2\theta)\right]$$

$$= \left(\frac{\cos 2\theta}{2}\right)_{\theta=0}^{2\pi} + \frac{1}{2}\left[\theta - \frac{\sin 2\theta}{2}\right]_{\theta=0}^{2\pi}$$

$$= \frac{1}{2}\left[\cos 4\pi - \cos 0\right] + \frac{1}{2}\left[\left(2\pi - \frac{\sin 4\pi}{2}\right) - \left(0 - \frac{\sin 0}{2}\right)\right]$$

$$= 0 + \frac{1}{2}(2\pi - 0) - (0-0)$$

$$\therefore \oint_C \bar{F}\cdot d\bar{r} = \pi$$

To evaluate surface integral

$$\text{curl}\,\bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix}$$

$$= \bar{i}\left(\frac{\partial}{\partial y}(-y^2z) - \frac{\partial}{\partial z}(-yz^2)\right) - \bar{j}\left(\frac{\partial}{\partial x}(-y^2z) - \frac{\partial}{\partial z}(2x-y)\right)$$
$$+ \bar{k}\left(\frac{\partial}{\partial x}(-yz^2) - \frac{\partial}{\partial y}(2x-y)\right)$$

$$= \bar{i}\left[-2yz + 2yz\right] - \bar{j}\left[0-0\right] + \bar{k}\left[0+1\right]$$

$$\text{curl}\,\bar{F} = \bar{k}$$

$$\therefore \text{curl}\,\bar{F}\cdot\bar{n} = \bar{k}\cdot\bar{k} = 1 \qquad (\because \bar{n} = \bar{k} \text{ for xy plane})$$

$$\therefore \int_S \text{curl}\,\bar{F}\cdot\bar{n}\,ds = \iint_R dx\,dy$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{1} r\,dr\,d\theta$$

$x = r\cos\theta$
$y = r\sin\theta$
$dx\,dy = r\,dr\,d\theta$

$$= \int_{\theta=0}^{2\pi} \left[\frac{r^2}{2}\right]_0^1 d\theta$$

$$= \int_{\theta=0}^{2\pi} \frac{1}{2}\, d\theta$$

$$= \frac{1}{2}\left[\theta\right]_0^{2\pi}$$

$$= \frac{1}{2}(2\pi)$$

$$\int_S \operatorname{curl} \bar{F} \cdot \bar{n}\, ds = \pi.$$

$$\therefore \oint_C \bar{F} \cdot d\bar{r} = \int_S \operatorname{curl} \bar{F} \cdot \bar{n}\, ds$$

Hence stoke's theorem verified.

11) verify Gauss Divergence theorem for $\bar{F} = (x^2 - yz)\bar{i} + (y^2 - zx)\bar{j} + (z^2 - xy)\bar{k}$ taken over the rectangular parallelepiped $0 \le x \le a$, $0 \le y \le b$, $0 \le z \le c$.

A) Given diversions theorem we have

$$\int_V \operatorname{div}\bar{F}\, dV = \int_S \bar{F}\cdot\bar{n}\, ds$$

Given that $\bar{F} = (x^2-yz)\bar{i} + (y^2\cancel{z} - zx)\bar{j} + (z^2-xy)\bar{k}$

$$= f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$$

$$\therefore \operatorname{div}\bar{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= 2x + 2y + 2z$$

$$\operatorname{div}\bar{F} = 2(x+y+z)$$

To evaluate the volume integral

In the given region 'z' varries from 0 to c, 'y' varries from 0 to b & 'x' varries from 0 to a

$$\therefore \int_V \operatorname{div}\bar{F}\, dV$$

$$= 2\int_{x=0}^{a}\int_{y=0}^{b}\int_{z=0}^{c} (x+y+z)\, dz\, dy\, dx$$

$$= 2\int_{x=0}^{a}\int_{y=0}^{b} \left(xz + yz + \frac{z^2}{2}\right)\Big]_{z=0}^{c} dy\, dx$$

$$= 2\int_{x=0}^{a}\int_{y=0}^{b} \left(xc + yc + \frac{c^2}{2}\right) dy\, dx$$

$$= 2\int_{x=0}^{a} \left(xyc + \frac{y^2}{2}c + y\frac{c^2}{2}\right)\Big]_{y=0}^{b} dx$$

$$= 2\int_{x=0}^{a}\left[xbc + \frac{b^2}{2}c + b\frac{c^2}{2}\right]dx$$

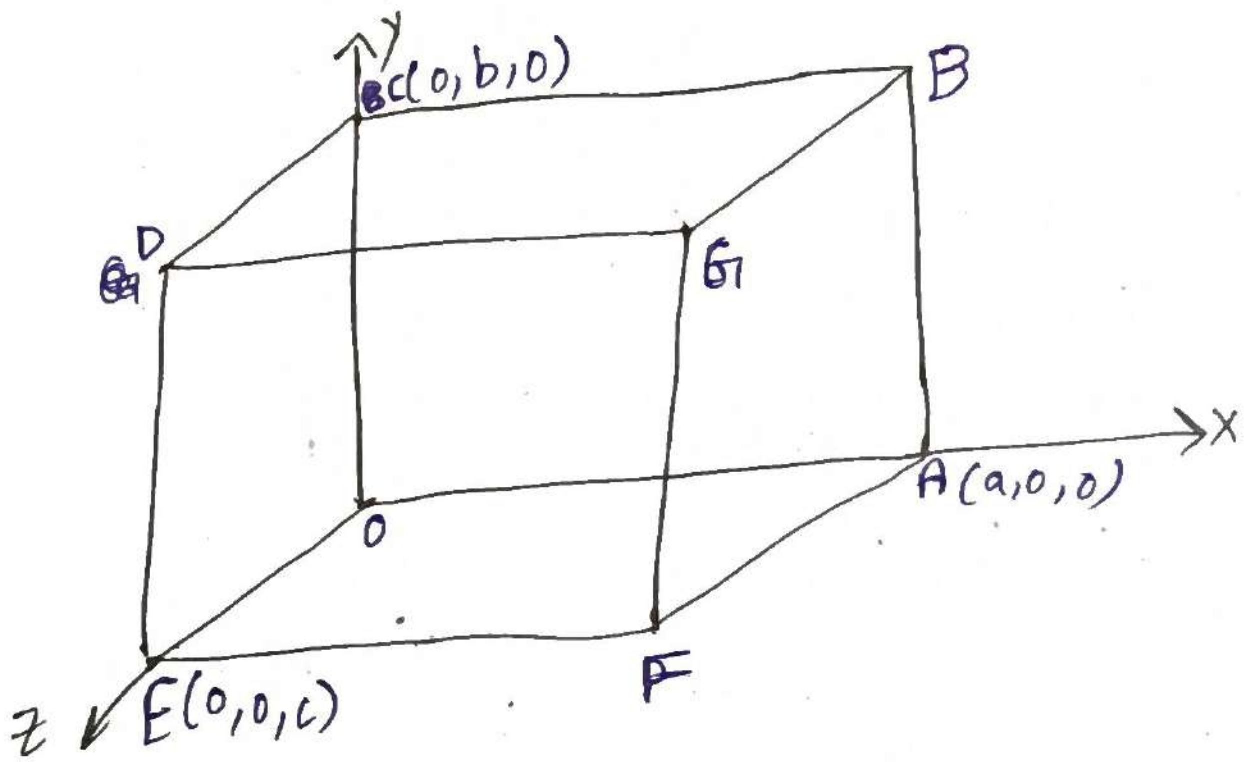
$$= 2\left[\frac{x^2}{2}bc + x\frac{b^2}{2}c + x\frac{bc^2}{2}\right]_{x=0}^{a}$$

$$= 2\left[\frac{a^2}{2}bc + a\frac{b^2}{2}c + a\frac{bc^2}{2}\right]$$

$$\int_V \operatorname{div}\bar{F}\,dv = a^2bc + ab^2c + abc^2$$

<u>To evaluate surface integral</u>

given that 's' is the surface over the rectangular parallelepiped $x=0$ $x=a$, $y=0$, $y=b$, $z=0$, & $z=c$



$$\therefore \int_S \bar{F}\cdot\bar{n}\,ds = \int_{S1}\bar{F}\cdot\bar{n}\,ds + \int_{S2}\bar{F}\cdot\bar{n}\,ds + \dots + \int_{S6}\bar{F}\cdot\bar{n}\,ds$$

where S_1 is the plane OABC, S_2 — DEFG, S_3 - OCDE, S_4 - ABGF, S_5 - OAFE, S_6 - BCDG

S.NO	plane	eq. of the plane	unit vector $\bar{n}$	$\bar{F}\cdot\bar{n}=(x^2-yz)\bar{i}+(y^2-zx)\bar{j}+(z^2-xy)\bar{k}$	ds	limits
1.	S1: OABC	$z=0$	$-\bar{k}$	$-z^2+xy$; $\therefore xy=0$; $\because z=0$	$dx\,dy$	$x\to 0$ to a, $y\to 0$ to b
2.	S2: DEFG	$z=c$	$\bar{k}$	$z^2-xy = c^2-xy$ $\because z=c$	$dx\,dy$	$x\to 0$ to a, $y\to 0$ to b
3.	S3: OCDE	$x=0$	$-\bar{i}$	$-x^2+yz$ $\because x=0$; yz	$dy\,dz$	$y\to 0$ to b, $z\to 0$ to c
4.	S4: ABGF	$x=a$	$\bar{i}$	x^2-yz $\because x=a$; a^2-yz	$dy\,dz$	$y\to 0$ to b, $z\to 0$ to c
5.	S5: OAFE	$y=0$	$-\bar{j}$	$-y^2+zx$ $\because y=0$; zx	$dx\,dz$	$x\to 0$ to a, $z\to 0$ to c
6.	S6: BCDG	$y=b$	$\bar{j}$	$y^2-zx = b^2-zx$ $\because y=b$	$dx\,dz$	$x\to 0$ to a, $z\to 0$ to c

i) along S1:-

$$\int_{S_1}\bar{F}\cdot\bar{n}\,ds=\int_{x=0}^{a}\int_{y=0}^{b}xy\,dy\,dx$$

$$=\int_{x=0}^{a}\left[\frac{xy^2}{2}\right]_{y=0}^{b}dx$$

$$=\int_{x=0}^{a}\frac{xb^2}{2}\,dx$$

$$\Rightarrow\left[\frac{x^2b^2}{4}\right]_{x=0}^{a}=\frac{a^2b^2}{4}$$

ii) along S_2 :-

$$\int_{S_2} \bar{F}\cdot\bar{n}\, ds = \int_{y=0}^{b}\int_{x=0}^{a} (c^2 - xy)\, dx\, dy$$

$$= \int_{y=0}^{b} \left(c^2 x - \frac{x^2}{2} y\right)\Big]_{x=0}^{a} dy$$

$$= \int_{y=0}^{b} \left[c^2 a - \frac{a^2}{2} y\right] dy$$

$$= c^2 a y - \frac{a^2}{4} y^2 \Big]_{y=0}^{b}$$

$$\int_{S_2} \bar{F}\cdot\bar{n}\, ds = abc^2 - \frac{a^2 b^2}{4}$$

iii) along S_3 :-

$$\int_{S_3} \bar{F}\cdot\bar{n}\, ds = \int_{z=0}^{c}\int_{y=0}^{b} yz\, dy\, dz$$

$$= \int_{z=0}^{c} \left(\frac{y^2}{2} z\right)\Big]_{y=0}^{b} dz$$

$$= \int_{z=0}^{c} \left[\frac{b^2}{2} z\right] dz$$

$$= \frac{b^2}{4} z^2 \Big]_{z=0}^{c}$$

$$\int_{S_3} \bar{F}\cdot\bar{n}\, ds = \frac{b^2 c^2}{4}$$

(14)

iv) along S_4:-

$$\int_{S_4} \overline{F}\cdot\overline{n}\, ds = \int_{z=0}^{c}\int_{y=0}^{b} (a^2 - yz)\, dy\, dz$$

$$= \int_{z=0}^{c} \left[a^2 y - \frac{y^2}{2} z\right]_{y=0}^{b} dz$$

$$= \int_{z=0}^{c} a^2 b - \frac{b^2}{2} z \;\, dz$$

$$= a^2 b z - \frac{b^2}{4} z^2 \Big]_{z=0}^{c}$$

$$\int_{S_4} \overline{F}\cdot\overline{n}\, ds = a^2 bc - \frac{b^2 c^2}{4}$$

v) along S_5:-

$$\int_{S_5} \overline{F}\cdot\overline{n}\, ds = \int_{z=0}^{c}\int_{x=0}^{a} zx\, dx\, dz$$

$$= \int_{z=0}^{c} \left[z \frac{x^2}{2}\right]_{x=0}^{a} dz$$

$$= \int_{z=0}^{c} z \frac{a^2}{2}\, dz$$

$$= \frac{a^2 z^2}{4} \Big]_{z=0}^{c}$$

$$\int_{S_5} \overline{F}\cdot\overline{n}\, ds = \frac{a^2 c^2}{4}$$

vi) along S_6 :-

$$\iint_{S_6} \bar{F}\cdot\bar{n}\, dS = \int_{z=0}^{c}\int_{x=0}^{a} z^2 - xb \, dx\, dz$$

$$= \int_{z=0}^{c} z^2x - \frac{x^2}{2}b\Big]_{x=0}^{a} dz$$

$$= \int_{z=0}^{c}\left(z^2a - \frac{a^2b}{2}\right)dz$$

$$= z^2$$

$$\int_{S_6} \bar{F}\cdot\bar{n}\, ds = \int_{z=0}^{c}\int_{x=0}^{a} (b^2 - zx)\, dx\, dz$$

$$= \int_{z=0}^{c} b^2x - \frac{zx^2}{2}\Big]_{x=0}^{a} dz$$

$$= \int_{z=0}^{c}\left[b^2a - \frac{za^2}{2}\right] dz$$

$$= b^2az - \frac{z^2a^2}{4}\Big]_{z=0}^{c}$$

$$\int_{S_6} \bar{F}\cdot\bar{n}\, ds = ab^2c - \frac{a^2c^2}{4}$$

$$\therefore \int_{S} \bar{F}\cdot\bar{n}\, ds = \int_{S_1} \bar{F}\cdot\bar{n}\, ds + \int_{S_2} \bar{F}\cdot\bar{n}\, ds + \dots + \int_{S_6} \bar{F}\cdot\bar{n}\, ds$$

(17)

$$\int_S \bar{F}\cdot\bar{n}\,ds = \cancel{\frac{a^2b^2}{4}} + abc^2 - \cancel{\frac{a^2b^2}{4}} + \cancel{\frac{b^2c^2}{4}} + a^2bc - \cancel{\frac{b^2c^2}{4}} + \cancel{\frac{a^2c^2}{4}} + ab^2c - \cancel{\frac{a^2c^2}{4}}$$

$$\int_S \bar{F}\cdot\bar{n}\,ds = a^2bc + ab^2c + abc^2$$

$$\int_S \bar{F}\cdot\bar{n}\,ds = abc(a+b+c)$$

$$\therefore \int_V \operatorname{div}\bar{F}\,dV = \int_S \bar{F}\cdot\bar{n}\,ds$$

hence Gauss divergence theorem verified.

12) Evaluate $\iint_S (x\,dy\,dz + y\,dz\,dx + z\,dx\,dy)$ by using divergence theorem, where S is the surface of the sphere $x^2+y^2+z^2=a^2$

A) Given $\iint_S x\,dy\,dz + y\,dz\,dx + z\,dx\,dy$

compare with $\iint_S F_1\,dy\,dz + F_2\,dx\,dz + F_3\,dx\,dy$

$$F_1 = x,\ F_2 = y,\ F_3 = z$$

$$F = F_1\bar{i} + F_2\bar{j} + F_3\bar{k} = x\bar{i} + y\bar{j} + z\bar{k}$$

$$\bar{\nabla}\cdot F = \left(\bar{i}\frac{\partial}{\partial x} + \bar{j}\frac{\partial}{\partial y} + \bar{k}\frac{\partial}{\partial z}\right)(x\bar{i} + y\bar{j} + z\bar{k})$$

$$= 1+1+1$$

$$\nabla\cdot\bar{F} = 3$$

Gauss divergence theorem states that

$$\int_S \bar{F}\cdot\bar{n}\,ds = \int_V \operatorname{div}\bar{F}\,dV$$

$$\iint F_1\,dy\,dz + \iint F_2\,dx\,dz + \iint F_3\,dx\,dy = \int_V 3\,dV$$

$$\iint x\,dy\,dz + y\,dx\,dz + z\,dx\,dy = 3\int_V 1\,dV$$

$$= 3V$$

$$= 3(\text{Volume of sphere } x^2+y^2+z^2=a^2)$$

$$= 3\times\frac{4}{3}\pi a^3$$

$$= 4\pi a^3$$

13) State the following vector integral theorems:

i) Green's theorem

ii) Stokes theorem

iii) Gauss divergence theorem.

A) i) Green's theorem

If a "R" is a closed region in xy-plane bounded by a simple closed curve "c" & if "M" & "N" are continous functions x & y having continous derivates in "R" then $\boxed{\oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy}$

where "c" is traversed in positive direction.

ii) Stoke's theorem

Let 's' be a open surface bounded by a closed non-intersecting curve 'c' if F is any differentiable vector point function then

$$\boxed{\oint_C \bar{F} \cdot d\bar{r} = \int_S curl F \cdot \bar{n}\, ds}$$ where 'c' is traversed in positive direction & $\bar{n}$ is the outward drawn unit normal vector at any point of S.

iii) Gauss Divergence Theorem

Let 's' be a closed surface enclosing a volume 'v' if F is a continously differentiable vector point function then

$$\int_V div \bar{F}\, dv = \int_S F \cdot \bar{n}\, ds$$

where $\bar{n}$ is the outward drawn unit normal vector at any point of 'S'.